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Chromatin priming and Hunchback recruitment integrate spatial and temporal cues in *Drosophila* neuroblasts

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eLife Assessment

The study provides an **important** advance towards understanding how spatial and temporal transcriptional programs are integrated to regulate lineage-specific chromatin and enhancer activation. The functional evidence is currently **incomplete**, but the current data provide a **solid** correlative and conceptual foundation. Functional experiments directly linking Gsb occupancy to chromatin state and regulation of some lineage-specific targets would further strengthen the causal interpretation of the model. Clarifying the scope of conclusions and explicitly acknowledging the technical limitations of current chromatin assays would provide a more balanced interpretation of the manuscript.

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Abstract

Neural stem cells generate diverse cell types by integrating spatial and temporal cues to activate neuron-specific terminal selector (TS) genes. In *Drosophila* neuroblasts (NBs), spatial patterning sets lineage identity, while a temporal transcription factor (TTF) cascade sets birth order. Two proposed mechanisms could integrate these inputs. In *direct regulation*, spatial transcription factors (STFs) and TTFs co-occupy and activate TS enhancers within NBs. In *epigenetic regulation*, STFs first prime NB-specific chromatin, creating ‘sites of integration’ (SoIs) that later recruit TTFs.

We test this in two identified NBs — NB5-6 and NB7-4 — and their candidate STFs, Gooseberry (Gsb) and Engrailed (En), together with the first TTF, Hunchback (Hb). In NB5-6, Gsb is expressed transiently, suggesting a chromatin-based memory of its activity. In NB7-4, En expression persists throughout development so integration could either be epigenetic or direct. We used chromatin engagement by the STFs as the discriminator between these models. If integration is epigenetic, the STF must engage less-accessible chromatin to establish NB-specific SoIs; if regulation is direct, the STF need not.

We find that En binds only to pre-accessible loci in NB7-4 and En+Hb co-binding marks the most accessible enhancers. This suggests that NB7-4 likely relies on an unknown priming factor to establish SoIs, with direct En–Hb co-binding mediating enhancer activation.

In NB5-6, Gsb binds both open and less-accessible chromatin and Gsb+Hb co-binding marks the most accessible enhancers. When ectopically expressed, Gsb remodels chromatin globally in the non-cognate NB7-4, and at endogenous NB7-4 SoIs, it specifically reduces accessibility as well as Hb binding. This suggests that in NB5-6 Gsb likely acts together with other NB5-6-specific factors to recognize less-accessible chromatin and to promote Hb recruitment while restricting Hb occupancy to appropriate enhancers. Together these findings support a unified two-step model: NB-specific combinations of TFs — each NB’s “STF code” — first prime chromatin and then recruit and restrict Hb to ensure lineage-specific enhancer activation.

Introduction

The diversity of cell types in the nervous system arises from a small pool of neural stem cells (NSCs). Decades of work in *Drosophila* show that NSCs use a hierarchical regulatory logic: positional cues endow NSCs with distinct molecular identities and lineage potentials, while a temporal gene cascade determines the production of different neuron types over time. NSCs activate neuron-specific terminal selector (TS) genes by integrating these spatial and temporal inputs along with Notch signalling. TSs then drive effector gene programs that implement neuronal morphology, connectivity, and neurotransmitter identity (1–3).

This has been best elucidated in the embryonic ventral nerve cord (VNC) and the optic lobe. In the VNC, 30 neuroblasts (NBs) per hemisegment each generate a lineage with a unique identity that is determined at the time of delamination (4–8). This identity reflects the NB's neuroectodermal origin, where intersecting anterior–posterior (A–P) and dorsoventral (D–V) patterning systems create distinct domains of TF expression (2, 3, 9) (Fig. 1A,B). NBs inherit these spatial TFs, which specify lineage identity. For example, the A–P gene *gsb* is expressed in row 5 of the neuroectoderm and required for row-5 identity: its loss transforms NBs to a row 3/4 fate, while misexpression induces the converse transformation (10). Other STFs act non-autonomously: *wingless* (*wg*) shapes the identity of adjacent rows without directly specifying row 5 (11). More posteriorly, *en* defines rows 6 and 7: *en* mutants lack NB6-4 and NB7-3, while *en* misexpression duplicates NB7-3 identity (12–14). D–V patterning factors, including *vnd*, *ind*, and *msh*, similarly define medial, intermediate, and lateral NB columns. Gain- and loss-of-function studies demonstrate that these D–V genes also respecify NB columnar identity (15–18). Together, these intersecting spatial cues generate 30 distinct NB identities per hemisegment, each with a unique developmental potential.

In the optic lobe, an analogous spatial logic partitions the neuroepithelium of the outer and inner proliferation centers (OPC/IPC) into A–P domains by *Vsx*, *Optix*, and *Rx* (split by *Dpp* and *Wg* and into D–V compartments by *Spalt/Salm* and *Disco/Hh*) (1, 19–23). These factors establish progenitor territories that seed distinct neuronal outputs across the medulla, lobula, and lobula plate (20–23). For example, *Vsx*-domain progenitors generate Pm3 neurons, dorsal and ventral *Rx* territories produce Pm1 and Pm2 respectively, whereas *Optix*-domain equivalents are eliminated by apoptosis (20). As in the VNC, manipulating spatial inputs reassigns territories and shifts lineage output. For example, expanding the *Rx* domain biases neural fates towards Pm1/2 output, and mutating *disco*, which specifies ventral fates, results in loss of ventral medulla neuron identities (20, 22).

NBs are also patterned in time. Each NB sequentially expresses a cascade of TTFs, creating successive competence windows during which different neurons are born (Fig. 1C) (1, 2, 19, 24, 25). In the embryonic VNC, the canonical cascade proceeds through Hunchback (*Hb*) → Kruppel (*Kr*) → Pdm → Castor (*Cas*) → Grainyhead (*Grh*) (26–28). In the optic lobe, this cascade consists of *Hth* → *Opa* → *Erm* → *Ey* → *Hbn* → *Scro* → *Slp* → *D* → *BarH1* → *Tll* (29–34). While the identity of the temporal factors varies across systems — including the central brain, type II lineages, and in the postembryonic VNC and brain — the strategy is conserved: sequential expression of TTFs subdivides neurogenesis into discrete temporal windows, each producing neuron types appropriate to that time point.

In keeping with this, manipulating temporal factors respecifies the temporal identity of the neurons. In the optic lobe, loss of function of *Ey*, *Slp*, or *D* stalls the temporal sequence and the neurons from those windows are lost; misexpressing any one of them biases production toward that window's fates (31). For example, in the early *Hth*/*Bsh* window, *bsh* or *hth* loss eliminates Mi1 neurons, and *Bsh* overexpression induces Mi1-like neurons (31, 35, 36). Similarly, in the embryonic VNC, *Hb*, the first in the temporal cascade, is both necessary and sufficient for the earliest-born neurons in multiple NB lineages. Its misexpression prolongs early competence windows and induces supernumerary early-born neurons, while its loss abolishes these fates (26, 27, 37–40).

Importantly, the identity of a neuron from a given time window differs across lineages. For example, *Hb*⁺ neurons are: U1 in NB7-1, RP2 in NB4-2, EW1 in NB7-3, and CCAP in NB3-5 — implying that temporal information alone is insufficient to specify neuronal identity (24). Instead,

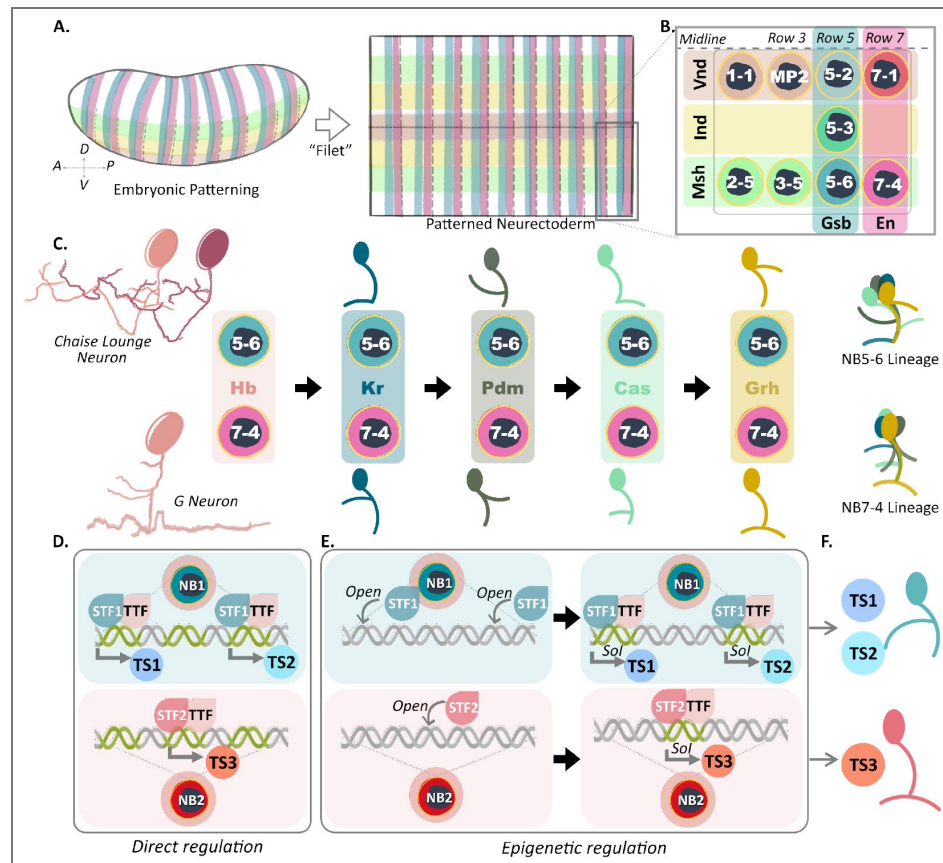


Fig. 1. Spatial and temporal patterning of neuroblasts and models for their integration.

A. Patterning the neuroectoderm: Early embryo patterning genes that establish the A-P and D-V body axes, also pattern the neuroectoderm, located ventro-laterally in the embryo. A 'filet' preparation allows visualization of the patterned neuroectoderm with representative A-P gene expression shown in blue and pink, and D-V columns in light green, yellow and red. **B. Spatial patterning:** NBs arise in an orthogonal array from this patterned neuroectoderm, each acquiring a unique molecular identity. **C. Temporal patterning.** Each spatially distinct NB (here, NB5-6 and NB7-4) transitions through a series of temporal windows corresponding to sequentially expressed TTFs, Hb > Kr > Pdm > Cas > Grh. This allows NBs to generate different neurons over time. During the Hb temporal window (pink box), NB5-6 generates Chaise Lounge neuron while NB7-4 generates G neuron. By the end of the temporal cascade, each NB produces distinct lineages. **D-F. Models for spatiotemporal integration.** Two NBs (NB1, NB2) express different spatial transcription factors (STF1, STF2) but share the same TTF. (D) Direct regulation: STFs and the TTF co-bind accessible enhancers of neuron type-specific terminal selector (TS) genes in each NB. (E) Epigenetic regulation: STFs first bind and open chromatin in an NB-specific manner, creating sites of integration (SoIs) that are later bound by the TTF to activate TS enhancers. (F) Outcome: distinct TS combinations (TS1/TS2 in NB1, TS3 in NB2) specify lineage-specific, time-appropriate neurons.

Hb indeed any TTF — must act in a spatially defined context to specify neuronal fate.

The mechanism by which spatial and temporal cues are integrated to generate such context-dependent outcomes remains unclear. Two mechanistic models have been proposed. In **direct regulation**, STF and TTFs (Hb) act concurrently: both are present at the same time in the same NB, and co-occupy TS enhancers to drive activation. In **epigenetic regulation**, the action is sequential: an STF first engages the enhancers to establish an accessible state, and only later does the TTF bind and activate transcription (Fig. 1D [41](#)).

Regardless of the model, we refer to the cis-regulatory loci that integrate the STF and TTF inputs as “sites of integration” (SoIs). Operationally, SoIs are enhancer elements that: (i) are specific to a particular NB lineage, and (ii) receive input from both the lineage-defining STF(s) and the TTF. Under direct regulation, SoIs are already in a receptive state and show concurrent STF–TTF occupancy. Under epigenetic regulation, SoIs bear evidence of prior STF action that establishes lineage-specific competence for later TTF recruitment (Fig. 1D [41](#)).

Here, we test these models using the two well-characterized embryonic lineages — NB5-6 and NB7-4 — and their putative STFs, Gsb and En respectively. Engrailed (En) persists in NB7-4 whereas Gooseberry (Gsb) is transient and precedes Hunchback (Hb) in NB5-6, suggesting that NB5-6 must use epigenetic regulation, while either could operate in NB7-4. We therefore used chromatin engagement as the discriminator between direct and epigenetic integration. If integration is epigenetic, the lineage STF must engage less-accessible chromatin to establish NB-specific sites of integration (SoIs) — enhancers rendered competent for later TTF recruitment. If regulation is direct, the STF need not engage less-accessible chromatin; concurrent STF–TTF binding on pre-accessible enhancers suffices. Guided by these predictions, we asked whether En in NB7-4 shows evidence of priming and whether Gsb in NB5-6 engages and remodels less-accessible loci—and whether such priming licenses later Hb recruitment.

Prior work mapping Hb and chromatin state in identified NBs showed that Hb binds distinct, lineage-specific loci that coincide with NB-specific open chromatin (SoIs) — and that these differences track the distribution of the STF Gsb — supporting a priming step that licenses later Hb binding ([41](#)). However, in that study, Gsb occupancy was obtained from available whole embryo ChIP-seq data and so NB-specific occupancy was lacking. Here we profile STF occupancy and chromatin accessibility in each lineage *in vivo* and test the sufficiency of STFs to reprogram accessibility and guide Hb binding. In NB7-4, En binds only pre-accessible loci, ruling En out as the priming factor and aligning with *direct regulation* on a pre-opened template. In NB5-6, Gsb binds both open and less-accessible chromatin and, when misexpressed in NB7-4, remodels chromatin genome-wide: it directly reduces accessibility at its binding targets, including at endogenous NB7-4 SoIs, where Hb occupancy is correspondingly weakened. It also indirectly renders some sites more accessible and such opened sites rarely recruit Hb. Across both lineages, the most accessible SoIs are co-bound by the STF and Hb, consistent with cooperative enhancer activation once chromatin has been appropriately primed.

Together, our findings support a unified model in which spatio-temporal integration occurs in two steps: NB-specific STF combinations first prime lineage-specific chromatin and then, together with lineage-specific cofactors, recruit and restrict Hb to generate NB-specific enhancer landscapes. This, in turn, supports a model where lineage identity is determined by unique combinations of STFs—the ‘STF code’—whose members partition the molecular functions of chromatin priming and Hb patterning that confer spatial identity within a lineage.

Results

Generation of functional Dam:Gsb and Dam:En to determine NB-specific STF occupancy

To determine STF occupancy *in vivo* in an NB-specific manner, we used Targeted DamID (TaDa), which has been employed previously to measure the Hb TTF’s occupancy in neuroblasts (NBs) ([41](#)). We generated UAS-LT3-Dam:Gsb and UAS-LT3-Dam:En fly lines, in which an upstream mCherry

ORF ensures low expression of Dam:Gsb or Dam:En from a downstream ORF (see **Methods**, Fig. S1A). This configuration minimizes toxicity associated with high Dam levels (42).

We first confirmed that Dam:Gsb and Dam:En were not toxic to cells. Their ubiquitous expression under Da-Gal4 caused no embryonic lethality and no denticle belt defects. However, this lack of toxicity might reflect non-functional TFs when fused to Dam. We therefore tested the functionality of these fusion proteins. We performed whole-embryo Dam:Gsb profiling (Da-Gal4>UAS-LT3-Dam:Gsb) and compared it to published Gsb ChIP data (43). Three biological replicates were generated, alongside UAS-LT3-Dam controls (DamOnly), and processed as described (see **Methods**). Dam:Gsb and DamOnly replicates correlated strongly within their own sets but poorly with each other (Fig. S1B), confirming reproducibility. Dam:Gsb also reproducibly marked known Gsb targets (*wg*, *gsb-n*, *prd*) (Fig. S1C), and signals were enriched at Gsb-ChIP peaks but not at ChIP peaks of unrelated TFs (Fig. S1D). Dam:Gsb peaks should be enriched for Gsb binding motifs. Although a canonical Gsb motif is not well-defined, Gsb contains both homeodomain and paired DNA-binding domains (44). A *de novo* motif search revealed enrichment of a paired-homeobox composite motif in Dam:Gsb peaks relative to background (Fig. S1E). Similarly, Dam:En data were reproducible, showed occupancy at established En targets, and were enriched for the canonical En motif (Fig. S1F-H).

Together, these experiments confirm that Dam:Gsb and Dam:En retain DNA-binding activity, are non-toxic to cells, and accurately report genomic occupancy, providing validated tools to interrogate STF binding in NB5-6 and NB7-4.

Assaying spatial TF occupancy in identified neuroblasts

STFs such as Gsb and En are expressed broadly in the embryonic neuroectoderm, yet their lineage-specific activities are executed within individual NBs. This creates a major challenge for profiling their genomic occupancy: each NB comprises only a small number of cells, and STF expression is often transient. Previous studies relied on whole-embryo ChIP data (45), which cannot resolve NB-specific binding. To overcome this limitation, we combined TaDa with NB-specific Gal4 drivers to assay STF binding in two identified lineages: Dam:Gsb in NB5-6 and Dam:En in NB7-4 (Fig. 2A-B). We collected data during stages 10–12, corresponding to the Hb competence period when SoIs are active. Despite sampling very few cells over this short time window, replicate Dam:Gsb and Dam:En profiles were highly reproducible and detected occupancy at established Gsb and En target loci (Fig. 2C-F). Dam-only controls confirmed previous findings that NB5-6 and NB7-4 possess distinct chromatin landscapes, with many loci uniquely accessible in one NB but not the other (1933 loci at $FDR \leq 0.05$; Fig. 2G-H). Together, these results validate that TaDa can be used to reliably profile STF occupancy in individual NBs, providing the necessary tools to test how spatio-temporal integration occurs within neuroblasts.

En binds only accessible chromatin in NB7-4

The two models - *direct* or *epigenetic regulation* - make clear predictions for STF occupancy in the context of the chromatin within NBs. While STF occupancy at less accessible chromatin is necessary under the *epigenetic regulation*, it is dispensable under *direct regulation*.

To test whether NB7-4 follows direct or epigenetic regulation, we asked whether En binds only at pre-open sites (consistent with conventional TF behaviour) or also engages less accessible chromatin to create SoIs (as predicted by epigenetic priming).

We first focussed our analyses on the previously generated SoIs (41). These sites were identified by performing a differential analysis on Hb binding in NB5-6 and NB7-4, and were confirmed to be differentially accessible in the two NBs as previously reported (see **Methods**, Fig. S2, Fig. 3A). As expected, En was enriched at NB7-4-specific SoIs, where Hb uniquely binds in this lineage (Fig. 3B). However, there was no detectable En signal in NB7-4 at sites known to be NB5-6-specific SoIs (which remain inaccessible and Hb-free in NB7-4) (Fig. 3C). Thus, En associates with SoIs only once they are already open and competent for Hb binding, but does not engage closed SoIs that might require priming.

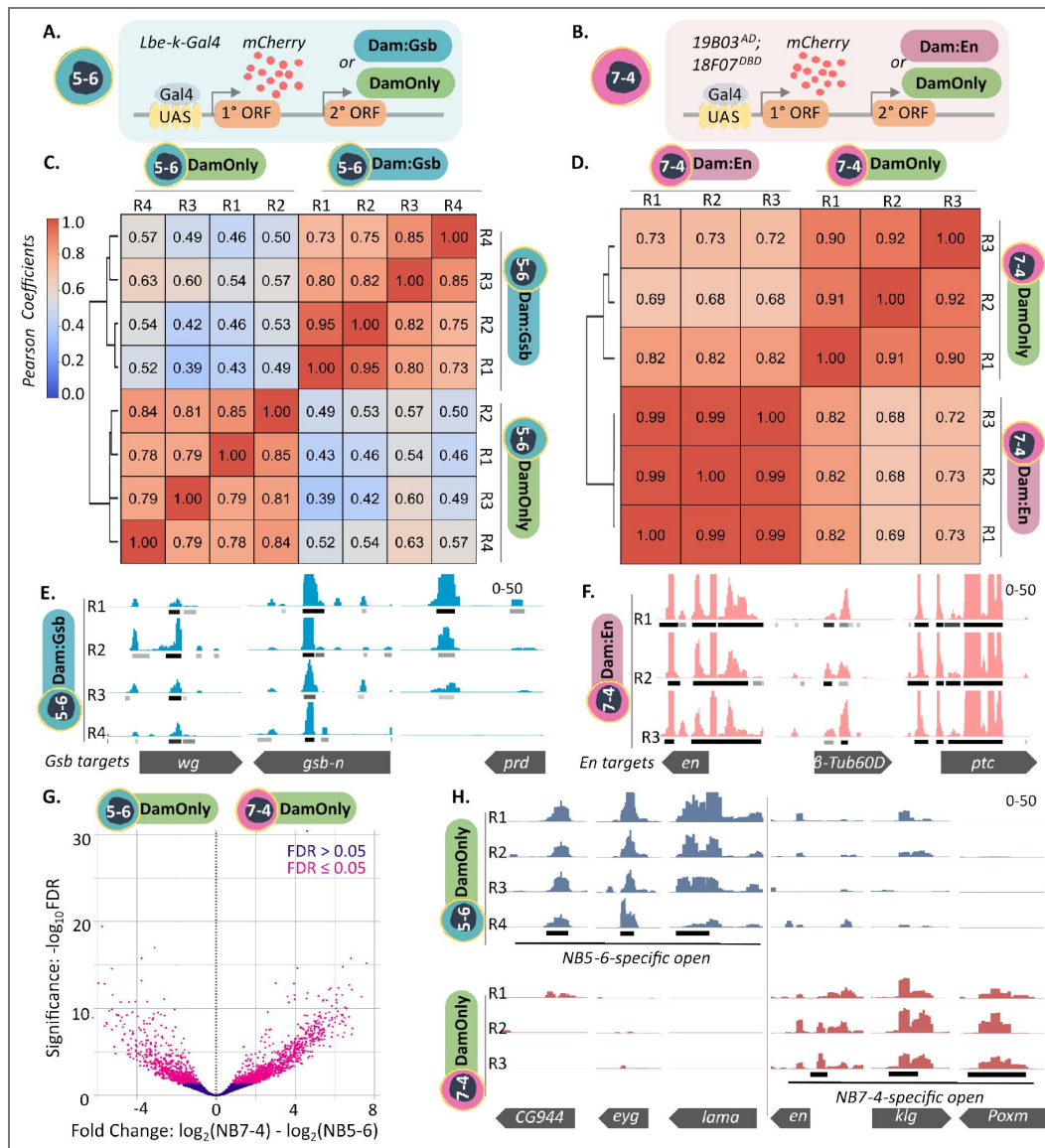


Fig. 2. NB5-6 Gsb and NB7-4 En TaDa profiling and lineage-specific chromatin accessibility.

A,B. TaDa strategy to profile NB5-6 Gsb and NB7-4 En occupancy and chromatin accessibility. In NB5-6, Lbe-k-Gal4 drives a UAS construct in which mCherry is encoded in the primary ORF and Dam:Gsb (blue) or DamOnly (green) in the secondary ORF. In NB7-4, the split-Gal4 line 19B03AD;18F07DBD similarly drives Dam:En (red) or DamOnly. High mCherry expression reports driver specificity, while translation of Dam fusions from the secondary ORF keeps Dam levels low, minimizing toxicity and preserving binding specificity. DamOnly profiles are used as a proxy for chromatin accessibility. **C. NB5-6-specific DamOnly and Dam:Gsb bind reproducibly.** Heatmap shows pairwise-correlation between NB5-6-specific DamOnly and Dam:Gsb replicates. Hierarchical clustering separates DamOnly and Dam:Gsb replicates, with higher Pearson correlation coefficient values within conditions than between them. **D. NB7-4-specific DamOnly and Dam:En bind reproducibly.** Heatmap shows pairwise correlations between NB7-4-specific DamOnly and Dam:En replicates, where hierarchical clustering separates DamOnly and Dam:En samples, with stronger correlations within than between conditions. **E. Binding at known Gsb targets.** Genome browser snapshots of 4 replicates of NB5-6-Dam:Gsb show binding at known Gsb targets- *wg*, *gsb-n* and *prd* (46–48). **F. Binding at known En targets:** Genome browser snapshots of 3 replicates of NB7-4-Dam:En show binding at *en*, *beta-Tub60D*, *ptc* (49, 50). **G. Chromatin is differentially accessible between NB5-6 and NB7-4.** Volcano plot showing sites differentially bound by DamOnly in NB5-6 and NB7-4. Differentially accessible loci with $\text{FDR} \leq 0.05$ are shown in magenta, while those with $\text{FDR} > 0.05$ are shown in blue. This threshold yielded 1933 differential sites. **H. Visualizing differentially accessible loci.** Genome browser snapshots show examples of differentially accessible sites in NB5-6 (blue tracks) and NB7-4 (red tracks). For all browser snapshots, Data range 0-50. Genotypes for NB5-6-Gsb occupancy: *w*; *Lbe-k-Gal4/+*; *+UAS-Dam:Gsb*. NB5-6-chromatin accessibility: *w*; *Lbe-k-Gal4/+*; *+UAS-DamOnly*. NB7-4-En occupancy: *w*; *19B03[AD]/+*; *18F07[DBD]/UAS-Dam:En*. NB7-4-chromatin accessibility: *w*; *19B03[AD]/+*; *18F07[DBD]/UAS-DamOnly*.

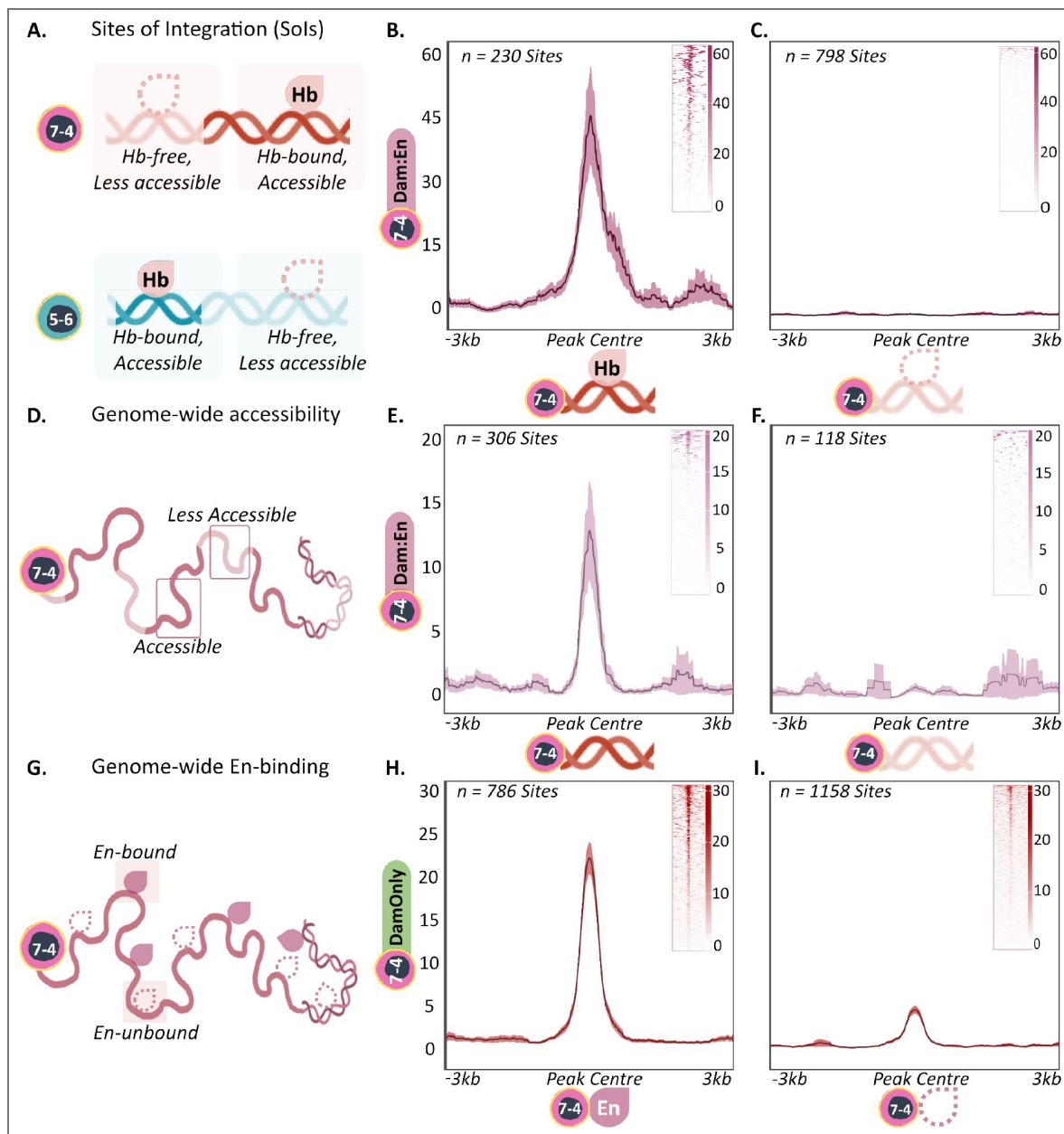


Fig. 3. En binds accessible chromatin.

A. En binding relative to Sites of Integration (SoIs). Top: Schematic representing NB7-4-SoIs, defined as loci that are Hb-bound (light pink drop) and accessible in NB7-4 (dark red DNA strand); loci corresponding to NB5-6-SoIs are Hb-free (hollow drop) and less accessible (light red DNA strand) in NB7-4. Bottom: NB5-6-SoIs are Hb-bound (light pink drop) and accessible in NB5-6 (dark blue DNA strand); loci corresponding to NB7-4-SoIs are Hb-free (hollow drop) and less accessible (light blue DNA strand) in NB7-4. Differential analysis identified 798 NB5-6-SoIs and 230 NB7-4-SoIs ($FDR \leq 0.01$, $FC \geq 2$, see Methods, Fig. S2). **B-C En binding at SoIs.** En binds NB7-4-SoIs in NB7-4 (**B**). No En binding is observed at NB5-6-SoIs in NB7-4 (**C**). Insets show heatmaps of En binding at corresponding loci, averaged across 3 replicates. **D-F. En binding relative to genome-wide chromatin accessibility.** Schematic shows uniquely accessible sites in NB7-4 (306 sites) in dark red, while those less accessible (118 sites) are shown in light red ($FDR \leq 0.01$, $FC \geq 2$, see Methods, Fig. S3). En binds accessible sites in NB7-4 (**E**). No En binding is detected at less accessible sites (**F**). Insets show heatmaps of En binding at the corresponding loci, averaged across 3 replicates. **G-I. Chromatin accessibility in NB7-4 relative to genome-wide En-bound and En-unbound sites.** Schematic representing En-bound sites (dark pink drops; 786 sites) and En-unbound sites (hollow drops; 1158 sites) ($FDR \leq 0.01$, $FC \geq 2$; see Methods, Fig. S4). Chromatin is accessible at En-bound sites in NB7-4 (**H**) and is less accessible at En-unbound sites (**I**). Insets show heatmaps of Dam signal from NB7-4 at the corresponding loci, averaged across 3 replicates. In all signal plots, solid lines indicate mean, coloured ribbons indicate 95% confidence interval. Genotypes for NB5-6-Hb occupancy: *w*; *Lbe-k-Gal4*/+; +/*UAS-Dam:Hb*. NB7-4-Hb occupancy: *w*; *19B03[AD]*/+; *18F07[DBD]/UAS-Dam:Hb* (41). NB7-4-En occupancy: *w*; *19B03[AD]*/+; *18F07[DBD]/UAS-Dam:En*. NB7-4-chromatin accessibility: *w*; *19B03[AD]*/+; *18F07[DBD]/UAS-DamOnly*.

This argues against En's role as an STF that primes lineage-specific chromatin by binding both accessible and less accessible regions to establish SoIs. Before concluding that En does not act via the epigenetic model, we examined its binding genome-wide, relative to chromatin accessibility (Fig. 3D). Because we had already validated the Dam:En tool and defined genome-wide NB-specific accessibility landscapes in NB7-4 and NB5-6 using Dam-only controls, we could confidently classify NB7-4 chromatin into loci that were uniquely accessible or less accessible through differential analysis of Dam binding between the two NBs: sites uniquely Dam-bound in NB7-4 were accessible and those that were uniquely open in NB5-6 were less accessible in NB7-4 (see **Methods**, Fig. S3, Fig. 3D).

At the genome-wide level, En was strongly enriched at accessible chromatin (Fig. 3E) and showed no detectable binding at less accessible sites (Fig. 3F). Thus, consistent with our earlier findings, En's binding is confined to pre-open regions. To test this relationship more directly, we compared chromatin accessibility at En-bound versus En-unbound loci genome-wide. A differential analysis between En (NB7-4) and Gsb (NB5-6) yielded 'En-bound' sites in NB7-4, that were not Gsb-bound in NB5-6. Conversely, 'En-unbound' sites corresponded to those that were Gsb-bound in NB5-6, but not En-bound in NB7-4 (see **Methods**, Fig. S4, Fig. 3G). En-bound regions displayed markedly higher accessibility than unbound ones (Fig. 3H-I), demonstrating that accessibility is not only correlated with but predictive of where En will bind. In other words, En binding is conditional on chromatin being pre-open. This pattern is consistent with conventional TF activity and argues against a pioneer role for En.

Taken together, these results show that En binds exclusively to accessible chromatin and associates with SoIs only after they are open, indicating that En does not act as a pioneer factor. Instead, it operates on enhancers that have already been primed - likely by other lineage-specific pioneer factors. This suggests that NB7-4 employs an 'STF code' in which different members have distinct functions — unknown pioneer factors establish its SoIs, while non-pioneer factors like En read the pre-open state and likely collaborate with Hb to activate lineage-specific enhancers (tested below).

Gsb binds both open and closed chromatin in NB5-6

Having established that En in NB7-4 binds pre-accessible chromatin, we next evaluated whether NB5-6 follows *direct* or *epigenetic regulation* by asking if Gsb, like En, binds pre-accessible chromatin or binds broadly across accessible and less accessible regions to establish SoIs (epigenetic priming). At NB5-6-specific SoIs (Fig. 4A), Gsb occupancy was enriched as expected (Fig. 4B). However, Gsb signal in NB5-6 was also enriched at sites identified to be NB7-4-specific SoIs, which remain less accessible and Hb-free in NB5-6 (Fig. 4C). Gsb occupancy at NB7-4-specific SoIs indicates that Gsb binding alone does not determine where Hb will be recruited.

This suggested that Gsb has the ability to bind less accessible chromatin. To confirm this, we extended our analysis to the whole genome and examined Gsb's occupancy in the context of chromatin accessibility. We used the previously generated Dam:Gsb binding data in NB5-6 and the already defined categories of chromatin loci from the previous sections - accessible in NB5-6 and less accessible in NB5-6. As in the previous section, differential analysis of Dam binding between the two NBs identified Dam-bound sites enriched specifically in NB5-6 as accessible, and NB7-4-enriched sites as less accessible in NB5-6 (see **Methods**, Fig. S3, Fig. 4D). At the genome-wide level, Gsb was enriched at accessible chromatin as expected (Fig. 4E). Importantly, we also detected some Gsb occupancy at less accessible regions (Fig. 4F). This property distinguishes Gsb from canonical TFs and is consistent with pioneer-like activity.

To confirm this, we compared chromatin accessibility at Gsb-bound versus unbound loci, which were determined as described earlier via differential analysis between Gsb (NB5-6) and En (NB7-4): enriched occupancy in NB5-6 yielded 'Gsb-bound' sites and 'Gsb-unbound' sites corresponded to those that were En-bound in NB7-4, but not Gsb-bound in NB5-6 (see **Methods**, Fig. S4, Fig. 4G). Genome-wide, accessibility was significantly lower at Gsb-bound regions relative to unbound ones (Fig. 4H-I), consistent with Gsb engaging both accessible and less accessible chromatin.

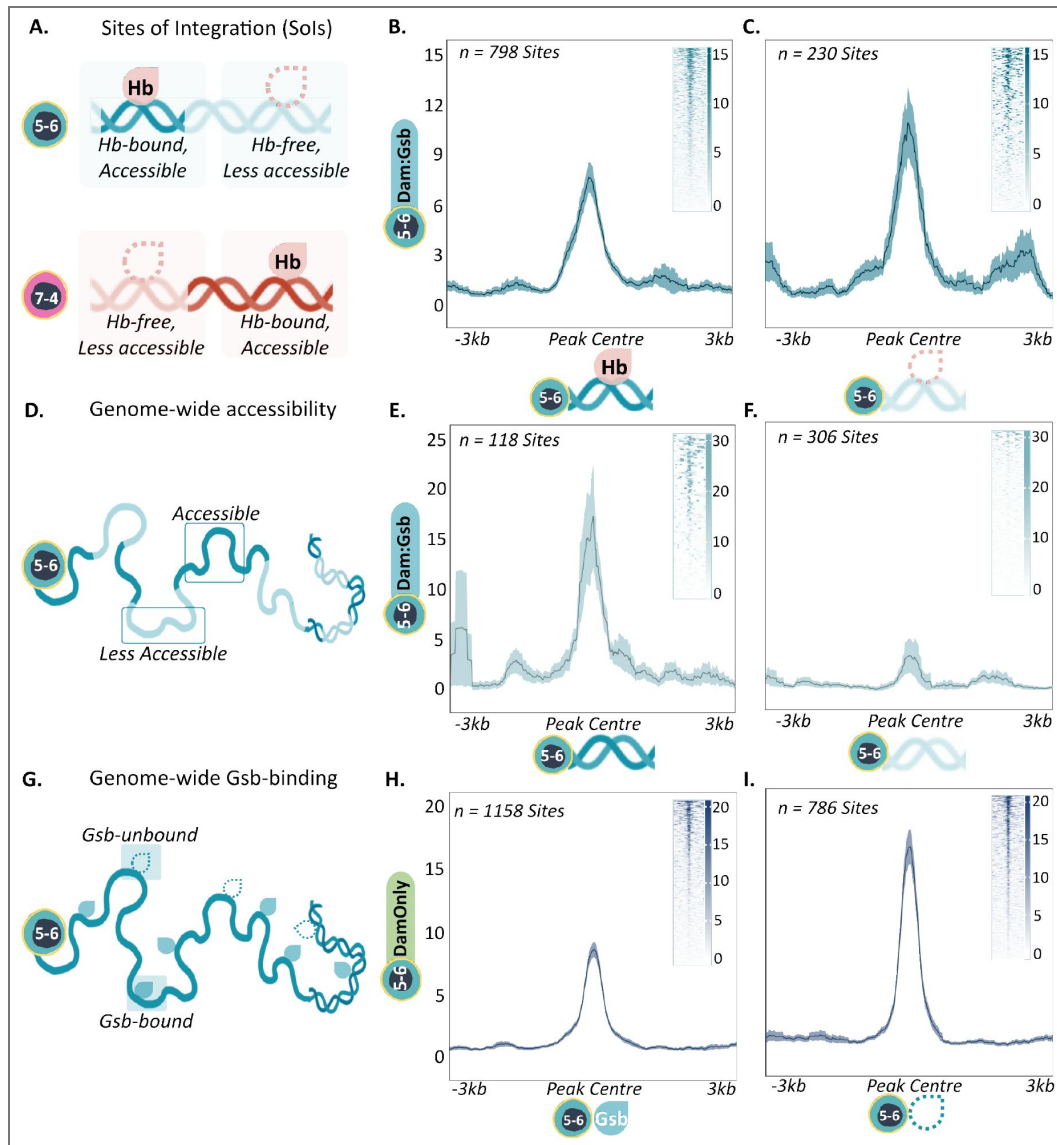


Fig. 4. Gsb binds both accessible and less accessible chromatin.

A. Gsb binding relative to Sites of Integration (SoIs). Top: Schematic representing NB5-6-SoIs, defined as loci that are Hb-bound (light pink drop) and accessible in NB5-6 (dark blue DNA strand); loci corresponding to NB7-4-SoIs are Hb-free (hollow drop) and less accessible (light blue DNA strand) in NB5-6. Bottom: NB7-4-SoIs are Hb-bound (light pink drop) and accessible in NB7-4 (dark red DNA strand); loci corresponding to NB5-6-SoIs are Hb-free (hollow drop) and less accessible (light red DNA strand). Differential analysis identified 798 NB5-6-SoIs and 230 NB7-4-SoIs ($FDR \leq 0.01$, $FC \geq 2$, see Methods, Fig. S2). Gsb binds NB5-6-SoIs (B) and at sites corresponding to NB7-4-SoIs, which are less accessible and Hb-free in NB5-6 (C). Insets show heatmaps of Gsb signal at the corresponding sites, averaged across 4 replicates. **D-F. Gsb binding relative to genome-wide chromatin accessibility.** Schematic represents DNA loci uniquely accessible (dark blue, 118 sites) and less accessible (light blue, 306 sites) in NB5-6 ($FDR \leq 0.01$, $FC \geq 2$, see Methods, Fig. S3). Gsb is bound at accessible sites in NB5-6 (E) and some Gsb binding is also seen at less accessible chromatin (F). Insets show heatmaps of Gsb signal at the corresponding loci, averaged across 4 replicates. **G-I. Chromatin accessibility in NB5-6 relative to Gsb-bound and Gsb-unbound sites.** Schematic representing Gsb-bound sites (blue drops, 1158 sites) and Gsb-unbound sites (hollow drops, 786 sites) ($FDR \leq 0.01$, $FC \geq 2$; see Methods, Fig. S4). Dam signal from NB5-6 mapped at Gsb-bound sites shows that chromatin is accessible at Gsb-bound sites in NB5-6 (H). At Gsb-unbound sites, chromatin is more accessible than at Gsb-bound loci (I). Insets show heatmaps of Dam signal from NB5-6 at the corresponding loci, averaged across 4 replicates. In all cases, solid lines indicate mean, coloured ribbons indicate 95% confidence interval. Genotypes for NB5-6-Hb occupancy: *w*; *Lbe-k-Gal4/+*; *+UAS-Dam:Hb*. NB7-4-Hb occupancy: *w*; *19B03[AD]/+*; *18F07[DBD]/UAS-Dam:Hb* (41). NB5-6-Gsb occupancy: *w*; *Lbe-k-Gal4/+*; *+UAS-Dam:Gsb*. NB5-6-chromatin accessibility: *w*; *Lbe-k-Gal4/+*; *+UAS-DamOnly*.

Together, these results meet a key prediction of the epigenetic model: Gsb binds both open and closed chromatin and occupies SoIs. However, because many Gsb-bound sites, including SoIs (those defined as NB7-4-specific), remain inaccessible and Hb-free in NB5-6, Gsb binding is not sufficient to recruit Hb. We infer that Hb recruitment in NB5-6 requires Gsb together with additional NB-specific cofactors rather than arising from Gsb–Hb direct co-occupancy alone. These findings also point to NB5-6 using an STF code with role partitioning: Gsb primes chromatin to establish SoIs, while other lineage-restricted factors recruit Hb and activate lineage-specific enhancers.

Co-binding with Hb marks highly accessible sites

Our analyses of Gsb in NB5-6 and En in NB7-4 suggest a division of labour within each lineage's STF code: some factors, like Gsb, may act as pioneers to prime chromatin, while others, like En, collaborate with Hb at enhancers that are already open. If enhancer activation requires both priming and TTF engagement, then loci co-bound by Hb and an STF-code component should represent the most active regulatory sites. To test this, we analyzed three genome-wide classes of loci — STF-only, Hb-only, and STF+Hb co-bound (Fig. 5A–B) — and compared their chromatin accessibility. For each TF (Gsb in NB5-6; En in NB7-4; Hb in both), the top 5000 highest-confidence peaks covering known genomic targets were used. To define STF-only sites, we subtracted Hb-bound peaks that overlapped with STFs; the same approach was used to obtain Hb-only sites in each NB. Co-bound sites were defined as those with an overlap of at least 90%. In both NB5-6 and NB7-4, accessibility was lowest at Hb-only sites, intermediate at STF-only sites, and highest at STF+Hb cobound loci (Fig. 5C–H). This analysis demonstrates that dual occupancy with Hb correlates with maximal accessibility at SoIs. This suggests that effective enhancer activation in these lineages proceeds in two steps: chromatin priming by STF-code components, then assembly of Hb with an STF-code partner. Co-binding marks the activated enhancers.

Gsb remodels chromatin when ectopically expressed

Having established that Gsb in NB5-6 can bind both open and closed chromatin, consistent with pioneer-like potential, we next asked whether Gsb is sufficient to actively remodel chromatin accessibility. Binding to closed loci is a hallmark of pioneers, but our proposed two-step model makes a stronger prediction: that an STF should not only recognize compact chromatin but also remodel it to generate accessible sites competent for later TTF binding and activation. Addressing this prediction requires moving beyond correlative occupancy analyses to a functional test of whether Gsb can alter chromatin state when introduced into a lineage where it is not normally expressed.

To this end, we ectopically expressed Gsb ubiquitously using a Hs:Gsb construct during the Hb-competence window and assayed chromatin accessibility in NB7-4, a lineage where Gsb is absent (see Methods, Fig. S5, Fig. 6A). This provided a context to assess whether Gsb could pioneer new sites of accessibility. Ectopic Gsb binding produced considerable remodelling: about 150 previously closed loci gained accessibility, while a similar number of normally open sites became less accessible (Fig. 6B,C). Crucially, only the decreasing sites overlap independent Gsb-binding datasets (whole-embryo Dam:Gsb), whereas the increasing sites do not (Fig. 6D,E). This pattern suggests that Gsb's direct effect is to reduce accessibility at its binding targets, while increases likely reflect indirect consequences of the perturbation.

We then asked whether such remodelling was sufficient to fulfil the second step of the model: recruitment of Hb to the newly accessible loci. To test this, we had simultaneously profiled Hb occupancy in the same embryos (see Methods). Despite Gsb-induced changes in chromatin accessibility, only a small fraction of these newly opened sites gained Hb binding, and even then the enrichment was weak (Fig. 6F,G).

We next focussed our analysis on established SoIs in NB7-4 and asked how ectopic Gsb affects them. At loci corresponding to NB5-6 SoIs, which are normally less accessible and Hb-free in NB7-4, Gsb misexpression neither increased accessibility nor recruited Hb: these sites remained closed and Hb-depleted (Fig. 6H,I). Thus, Gsb on its own is unable to install NB5-6-like SoIs in NB7-4. In

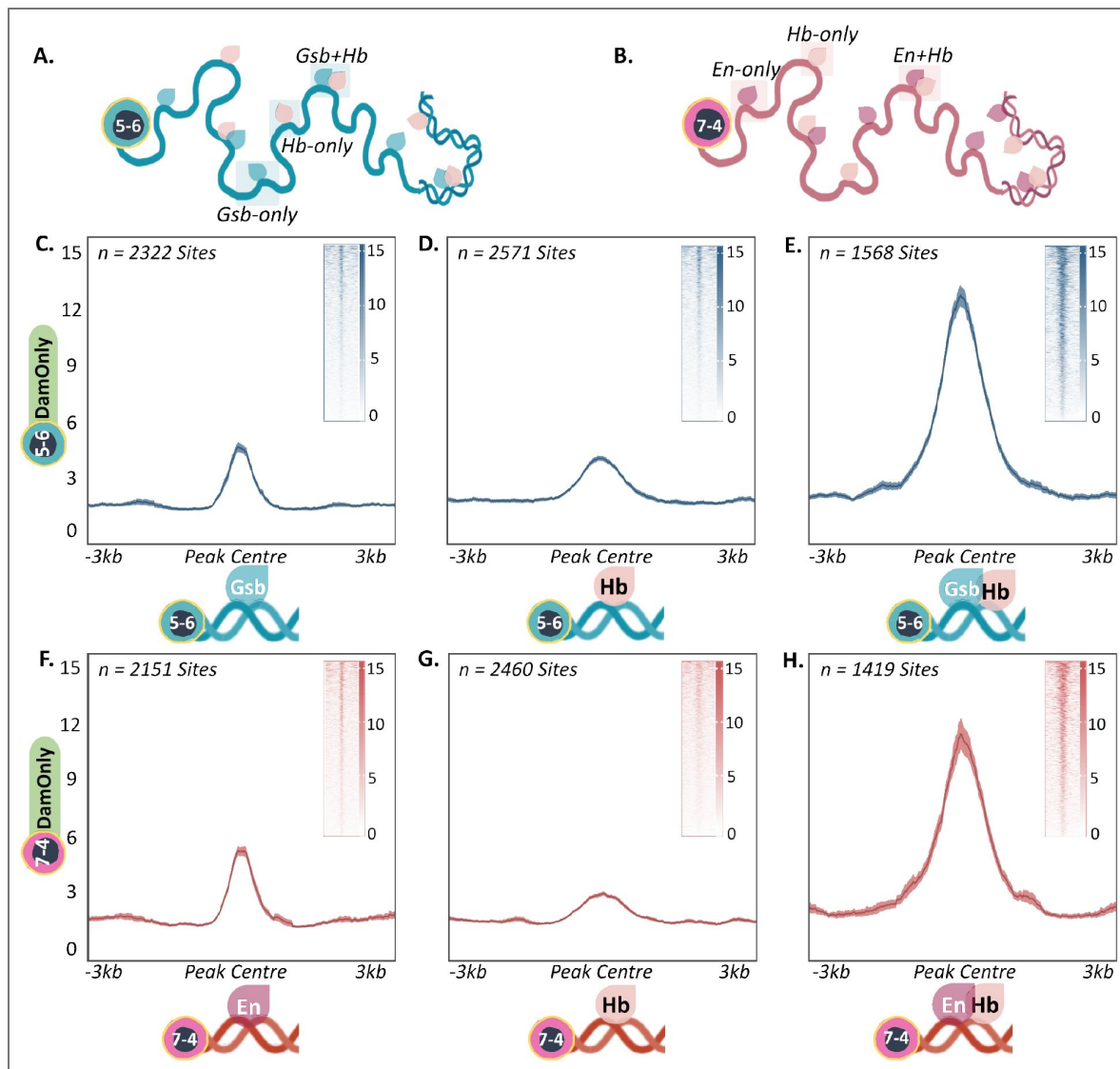


Fig. 5. Co-binding with Hb marks sites of highest accessibility.

Schematic in (A) shows sites in NB5-6 bound by Gsb only (blue drops, 2,322 sites), Hb only (light pink drops, 2,571 sites) and Gsb+Hb (1,568 sites). Co-bound sites were identified (see Methods). Schematic in (B) shows sites in NB7-4 bound by En only (dark pink drops, 2,151 sites), Hb only (light pink drops, 2,460 sites) and En+Hb (1,419 sites). Dam signal in NB5-6 at sites bound by Gsb only (C) and by Hb only (D) shows some accessibility, but Dam signal at Gsb+Hb co-bound shows most accessibility (E). Insets show heatmaps of NB5-6-Dam signal at the corresponding sites, averaged across 4 replicates. F. Dam signal in NB7-4 at sites bound by En only (F) and by Hb only (G) shows some accessibility, but Dam signal at En+Hb co-bound shows most accessibility (H). Insets show heatmaps of NB7-4-Dam signal at the corresponding sites, averaged across 3 replicates. Genotypes for NB5-6-Hb occupancy: *w*; *Lbe-k-Gal4/+*; *+UAS-Dam:Hb*. NB7-4-Hb occupancy: *w*; *19B03[AD]/+*; *18F07[DBD]/UAS-Dam:Hb* (41). NB5-6-Gsb occupancy: *w*; *Lbe-k-Gal4/+*; *+UAS-Dam:Gsb*. NB5-6-chromatin accessibility: *w*; *Lbe-k-Gal4/+*; *+UAS-DamOnly*. NB7-4-En occupancy: *w*; *19B03[AD]/+*; *18F07[DBD]/UAS-Dam:En*. NB7-4-chromatin accessibility: *w*; *19B03[AD]/+*; *18F07[DBD]/UAS-DamOnly*.

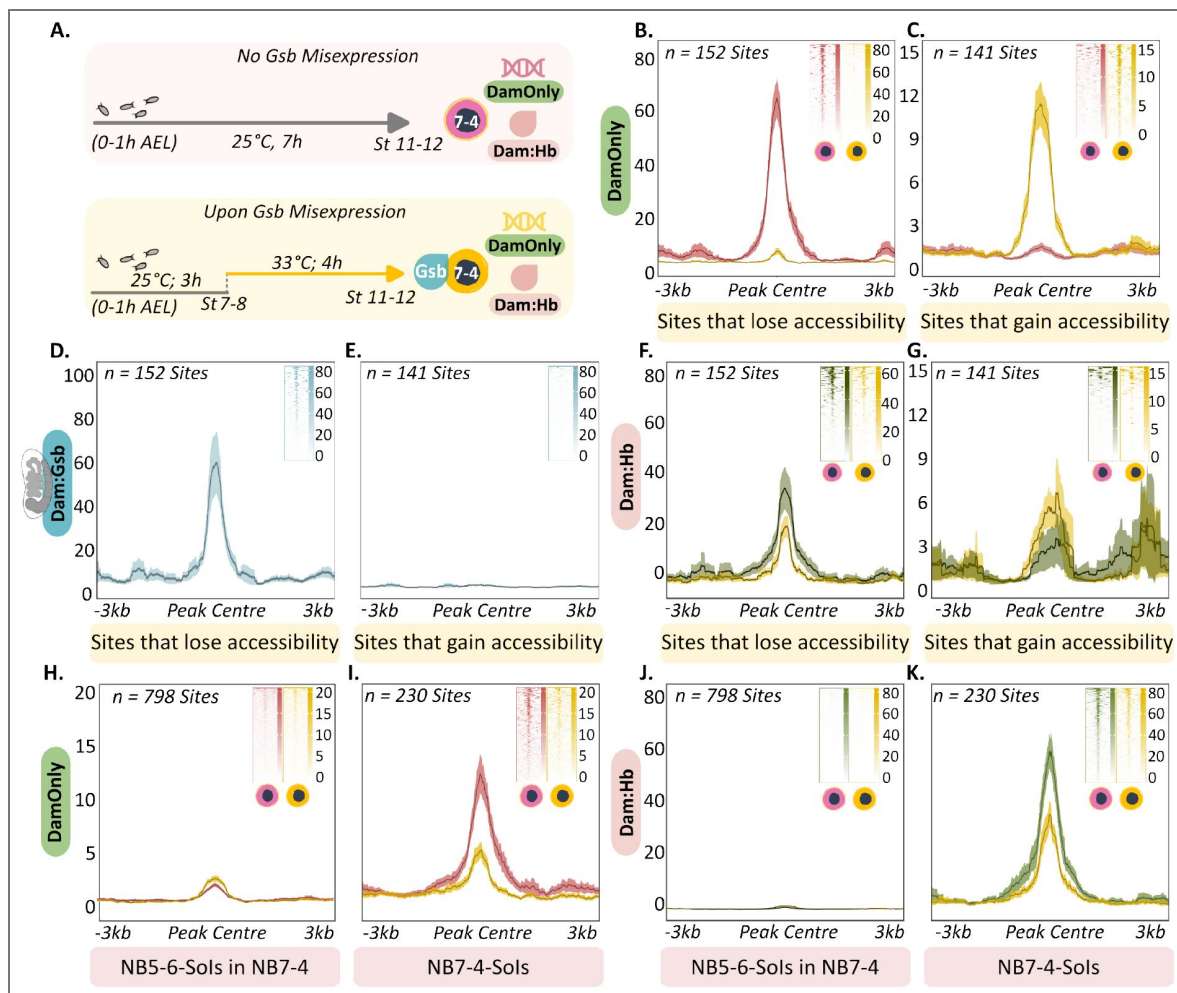


Fig. 6. Gsb remodels chromatin when ectopically expressed.

(A) Schematic of the experimental protocol while assaying chromatin accessibility and Hb occupancy (light pink drop) in control (top) and Gsb misexpressed (bottom) NB7-4. Control NB7-4 and its chromatin is denoted as a pink cell or DNA respectively, Gsb misexpressed NB7-4 and its chromatin is similarly denoted as yellow. **B-C. Chromatin accessibility changes in NB7-4 upon Gsb misexpression.** **B.** Loci that are accessible in control NB7-4 (red) and lose accessibility upon Gsb misexpression (yellow). **C.** Loci that are less accessible in control NB7-4 (red), but gain accessibility upon Gsb misexpression (yellow). Insets show heatmaps of control NB7-4-Dam (red) and Gsb misexpressed NB7-4-Dam (yellow) at the corresponding loci, averaged across 3 replicates each. **D-E. Whole embryo Gsb binding at the changed accessibility sites from B-C.** Whole embryo Gsb TaDa signal at sites corresponding to those that either lose **(D)** or gain **(E)** accessibility in NB7-4 upon Gsb misexpression. **F-G. Hb binding at the changed accessibility sites from B-C.** Hb signal in control (green) and Gsb misexpressed (yellow) NB7-4. Insets show heatmaps of control NB7-4-Hb (green) and with Gsb misexpression (yellow) at the corresponding loci, averaged across 3 replicates each. **H-I. Gsb mediated changes in chromatin accessibility at SoIs in NB7-4.** **H.** Chromatin in control (red) and Gsb misexpressed (yellow) NB7-4 corresponding to NB5-6-SoIs remains unchanged. **I.** Reduced chromatin accessibility in Gsb misexpressed NB7-4 (yellow) relative to control (red) at sites corresponding to NB7-4-SoIs. **J-K. Gsb mediated changes in Hb occupancy at SoIs in NB7-4.** **J.** Hb binding in control (green) and Gsb misexpressed (yellow) NB7-4 corresponding to NB5-6-SoIs remains unchanged. **K.** Reduced Hb binding in Gsb misexpressed NB7-4 (yellow) relative to control (green) at sites corresponding to NB7-4-SoIs. Genotypes for NB5-6-Hb occupancy: *w-; Lbe-k-Gal4/+; +/UAS-Dam:Hb*. NB7-4-Hb occupancy: *w-; 19B03[AD]/+; 18F07[DBD]/UAS-Dam:Hb* (41). NB5-6-Gsb occupancy: *w-; Lbe-k-Gal4/+; +/UAS-Dam:Gsb*. NB5-6-chromatin accessibility: *w-; Lbe-k-Gal4/+; +/UAS-DamOnly*. NB7-4-En occupancy: *w-; 19B03[AD]/+; 18F07[DBD]/UAS-Dam:En*. NB7-4-chromatin accessibility with Gsb misexpression: *hsGsb; 19B03[AD]/+; 18F07[DBD]/UAS-DamOnly*. NB7-4-Hb occupancy with Gsb misexpression: *hsGsb; 19B03[AD]/+; 18F07[DBD]/UAS-Dam:Hb*.

contrast, at endogenous NB7-4 SoIs, where chromatin is normally highly accessible and strongly bound by Hb, ectopic Gsb led to a marked reduction in accessibility (Fig. 6I). We also observed a reduction in Hb occupancy at these sites (Fig. 6K). Ectopic Gsb therefore partially antagonizes the native NB7-4 SoIs rather than creating new ones, consistent with its genome-wide role.

Together, these experiments demonstrate that Gsb is sufficient to remodel chromatin when misexpressed, providing direct functional evidence that it has pioneer-like activity. Importantly, our data suggest that Gsb acting on its own tends to close rather than open chromatin, including at SoIs. Furthermore, Gsb-induced accessibility changes only partially alter Hb binding, showing that enhancer activation requires more than chromatin opening. Taken together, these findings support and refine our earlier conclusion that productive opening and Hb recruitment arise from the NB-specific STF code, in which Gsb operates with additional lineage-specific factors to establish bona fide SoIs.

Manipulating Gsb alters NB lineage identity

Having shown that Gsb can function as a pioneer to remodel chromatin, we next asked whether this activity translates into changes in NB lineage identity. The two-step model predicts that pioneering activity by STFs establishes the chromatin landscape on which temporal factors act, thereby constraining lineage potential. If this is correct, then manipulating Gsb levels should directly impact NB fate.

We first depleted Gsb in the early embryo using Vasa-Cas9, which is known to be expressed in somatic cells of the early embryo (51), together with a *gsb*-targeted guideRNA. Embryos lacking *gsb* were embryonic lethal and exhibited axonal phenotypes consistent with previously described *gsb* loss (Fig. S6A-B). Sequencing such embryos at the time of NB delamination verified a mosaic but high rate of loss of Gsb function, confirming that we could induce *gsb* loss-of-function in early embryos (Fig. S6C-F).

In wild-type embryos, we could identify NB5-6-born Tv neurons, which express both Ap and Eya markers (52) (Fig. 7A, C-F). In *gsb* loss-of-function embryos, these NB5-6 markers along with the NB5-6-specific Lbe-K enhancer were lost, demonstrating that Gsb is necessary for maintaining NB5-6 identity (Fig. 7B, G-I). Ap-positive neurons are also born from MP2 (in row 3) and NB4-3, named dMP2 and dAp neurons, respectively (53) (Fig. 7A, C, J-L). We noted twice the usual number of dMP2 neurons in several hemisegments, while dAp neuron numbers remained unaffected (Fig. 7 M-O). In addition, there was a concomitant ectopic expression of the NB3-5-specific marker, Ems (Fig. 7P-Q). Since NB5-6-specific Eya- and Ap-positive cells arise only in the thoracic hemisegments, we limited our quantification of these markers to these hemisegments alone. Quantification of these markers across hemisegments confirmed this reciprocal fate switch - suppression of NB5-6 identity and gain of NB3-5 fate (Fig. 7R-T), demonstrating that Gsb is indispensable for maintaining NB5-6 identity and preventing re-specification toward an NB3-5 fate. These findings also confirm earlier reports that Gsb specifies row 5 and suppresses row 3 fates (10).

To test whether Gsb is sufficient to impose NB5-6 identity, we broadly misexpressed Gsb at the time of NB delamination and assayed the same set of molecular markers (Fig. 8A). Ectopic Gsb expression resulted in a loss of NB3-5 identity - a proportion of hemilineages showed a complete loss in NB3-5-reporter R59E09>GFP and Ems (Fig. 8B-C). Notably, Ems also marks NB3-3, another row 3 NB (6). Therefore, a complete loss of Ems from an entire hemisegment suggests a broad disruption of row 3 identity following ectopic Gsb expression, consistent with previous findings (10). Broad Gsb misexpression also induced a gain of NB5-6 identity, marked by Lbe>GFP. In several thoracic hemisegments, we observed supernumerary Eya- and Ap-positive neurons, consistent with a gain in NB5-6T-born Tv neurons (Fig. 8D-E). However, in only a small proportion of the hemisegments was there a concomitant gain in NB5-6-identity and loss of NB3-5 identity. Quantification confirmed these findings (Fig. 8F-J), indicating that while Gsb is sufficient to repress alternative fates, it is not always enough on its own to fully impose NB5-6 identity.

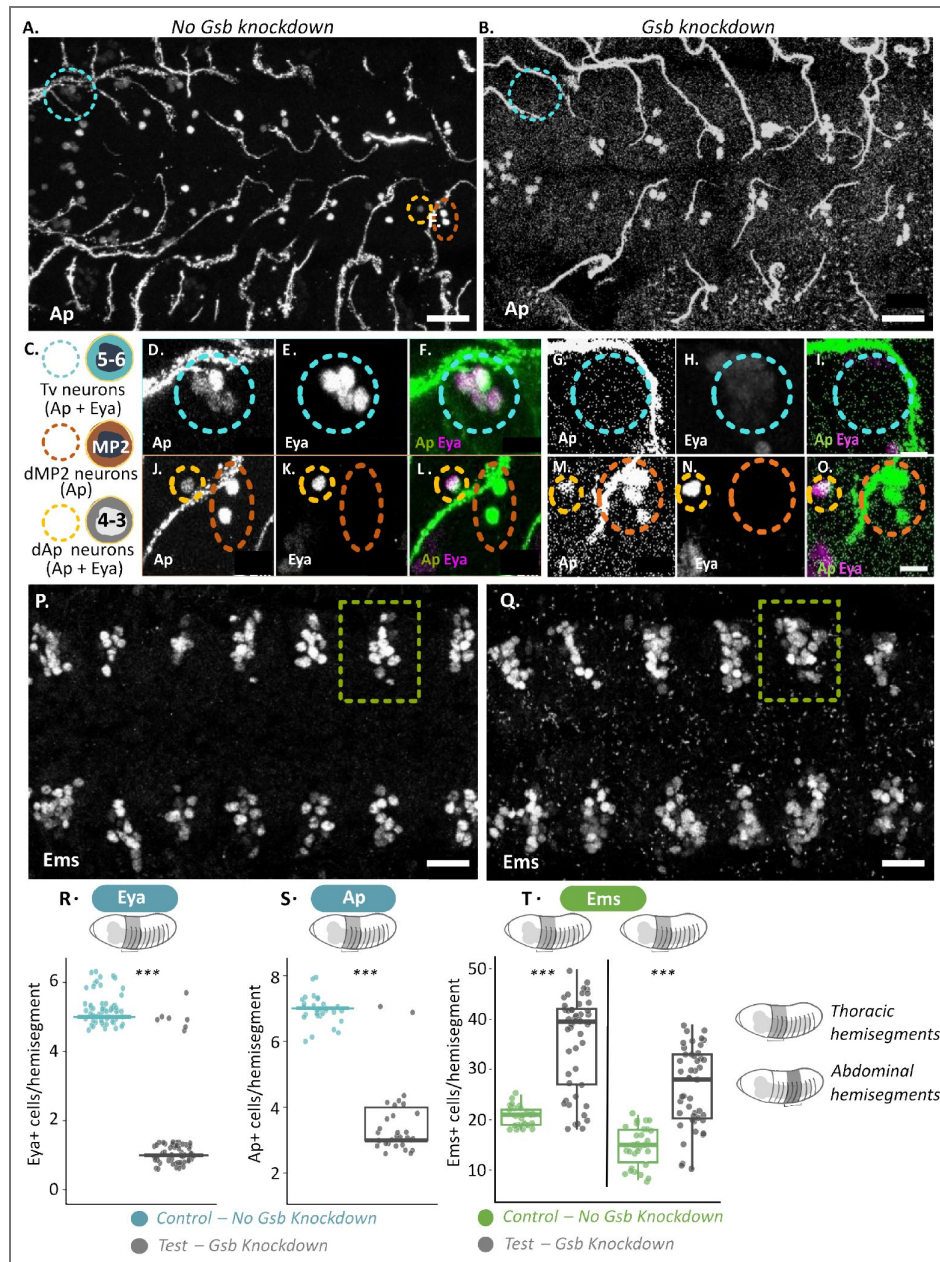


Fig. 7. Broad loss of *Gsb* function results in loss of NB5-6 identity and gain of NB3-5 identity.

A. Control embryo with no *Gsb* knockdown, showing Ap-positive neurons born from NB5-6 (cyan circle), MP2 (orange circle) and NB4-3 (yellow circle). **B.** Test embryos with *Gsb* knockdown show missing Ap-positive neurons from NB5-6 (cyan asterisks). Scale bar 10μ. **C.** Schematic summarizing Tv neurons (Ap+Eya), born from NB5-6, dMP2 neurons (Ap-positive) born from MP2, and dAp neurons (Ap+Eya) born from NB4-3. **D-F.** Tv neurons are a cluster of 4 Ap+ neurons born from NB5-6 in each thoracic hemisegment. These neurons express both Ap (D) and Eya (E). **G-I.** Upon *Gsb* knockdown, Tv neurons are absent (cyan asterisk). **J-L.** Every hemisegment has one dMP2 neuron born from MP2 (yellow dashed circle). There is also a pair of dAp neurons (orange dashed circle) born from NB4-3. dMP2 neuron also expresses Eya but dAp neurons do not. **M-O.** In test embryos with *Gsb* knockdown, dAP neuron (from NB4-3, yellow dashed circle) remains unaffected. However, there is an increase in the dMP2 neurons (orange dashed circle)-there are 2 pairs of Ap+ neurons here instead of one. Together (A-O) show a loss of NB5-6-like identity and a concomitant gain in MP2-like identity. NB4-3 lineage remains unaffected. Scale bar in insets 5μ. **P.** Control embryo showing Ems expression. **Q.** Upon broad knockdown of *Gsb*, there is an increase in Ems expression (green dashed box indicates an example of increased Ems). Scale bar 10μ. The decrease in thoracic Eya and Ap expression is quantified in (R) and (S). The gain in Ems expression has been quantified in (T). Each point represents a single hemisegment; boxplots indicate median and interquartile range. Statistical significance was assessed using two-tailed Mann-Whitney U tests. Significance levels are indicated as: *** p < 0.001, ** p < 0.01, * p < 0.05, ns = not significant. Genotype for control embryos: *Vasa Cas9/+*; For test embryos: *Vasa Cas9/+; gsb-guideRNA/+*.

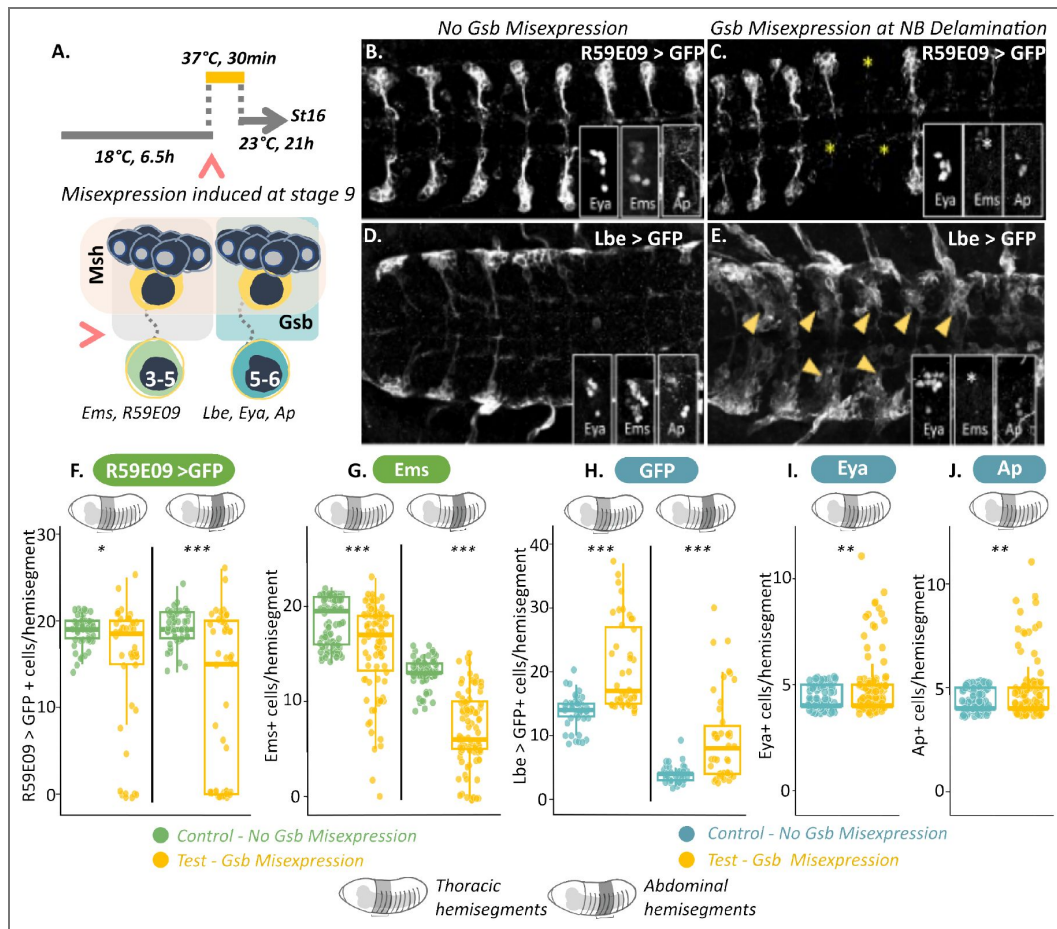


Fig. 8. Broad misexpression of Gsb results in loss of NB3-5 identity, but minimal gain of NB5-6 identity.

A. (Top) Schematic illustrating heat shock protocol used to induce broad Gsb misexpression at NB delamination (red arrowhead). (Bottom) Both NB3-5 and NB5-6 arise in the Msh domain; however, only NB5-6 normally expresses Gsb. NB3-5 identity is assessed using R59E09>GFP and Ems; NB5-6 identity is assessed using Lbe>GFP, Eya, and Ap. **B.** Control embryos with NB3-5 lineages labelled by R59E09>GFP. Insets show wild-type expression of Eya, Ems and Ap. **C.** Test embryos with broad Gsb misexpression shows some missing NB3-5 lineages (yellow asterisks). Insets show a concomitant loss of Ems (white asterisk); expression of Eya and Ap largely remains unaffected. **D.** Control embryos with NB5-6 lineage labelled by Lbe>GFP. Insets show wild-type expression of Eya, Ems and Ap. **E.** Test embryos with broad Gsb misexpression show ectopic Lbe>GFP expression (yellow arrowheads). Insets show a concomitant gain of Eya, but not Ap, and loss of Ems. The loss of NB3-5 identity is not always accompanied by a gain in NB5-6 identity. **(F–J)** Quantification of lineage markers under control vs test conditions. Each point represents a single hemisegment; boxplots show median and interquartile range. Quantification of GFP, Ems, Eya and Ap is shown separately in thoracic as well as abdominal segments. 4% of thoracic hemilineages showed gain of eya accompanied by loss of 3-5 lineage and its marker ems (n=48). 27% of thoracic and 38% of abdominal hemisegments showed loss of 3-5 lineages along with loss of ems (n=48). Statistical significance was assessed using two-tailed Mann-Whitney U tests. Significance levels are indicated as: *** p < 0.001, ** p < 0.01, * p < 0.05, ns = not significant. Genotypes for both control embryos (no heat shock) and test embryos (with heat shock): *hsGsb/w-; Lbe-k-Gal4, UAS-mCD8-GFP* and *hsGsb/w-; R59E09, UAS-myrGFP*.

Taken together, these experiments establish Gsb as the critical pioneer factor in establishing NB5-6 identity and suppressing NB3-5 fate. However, in both cases the transformations were incomplete: Gsb overexpression did not consistently install NB5-6 identity, and Gsb loss did not fully erase it. This parallel outcome suggests that while Gsb initiates chromatin remodelling and biases fate, complete lineage specification requires additional components of an NB5-6 STF code—factors that both stabilize identity and collaborate with temporal regulators like Hb.

Discussion

Neural stem cells must integrate spatial and temporal information to produce lineage-appropriate time-specific neurons. Two models have been proposed to explain how this could occur: **direct regulation** model, in which STF and TTFs cooccupy and activate enhancers of TS genes to deploy effector programs, and an **epigenetic regulation** model, in which early STFs establish open chromatin that later permits TTF binding and enhancer activation. Our results in two identified neuroblasts support a mechanism that contains features of both models. We propose that each NB carries a unique STF code — a combination of spatial TFs that confers lineage identity — whereby some members prime lineage-specific chromatin, while others collaborate with the TTF to recruit and activate lineage-specific enhancers.

The functional manipulations of *gsb* (both loss- and gain-of-function) are consistent with this division of labour: priming shifts chromatin accessibility and biases lineage identity, but full enhancer activation and therefore complete lineage transformation would require Hb recruitment by the appropriate NB-specific code. It is, however, possible that other neuroblasts or time windows use other strategies for integrating position and time — including TTF-led chromatin remodelling - the TTF Grainy head is a known pioneer (54) - or direct co-occupancy on DNA.

Gsb as a context-dependent pioneer

Our data firmly establish Gooseberry (Gsb) as a pioneer-like regulator in NB5-6. Genome-wide TaDa profiling shows that Gsb binds both accessible loci and regions that remain relatively inaccessible in this lineage. Ectopic expression experiments further demonstrate that Gsb is sufficient to remodel chromatin in the non-cognate NB7-4 in both directions: closing previously open sites and generating newly open ones. Together, these observations are consistent with context-dependent pioneer activity (55).

Direct pioneer activity has not been shown previously for *Drosophila* Gsb in neuroblasts, but Gsb is a Pax3/7-class TF that specifies row-5/6 NB identity in the neuroectoderm, placing it at the top of the spatial cascade that patterns these lineages (10). Within the Pax family, Gsb (and Prd) are Pax3/7 homologues by sequence and function (56, 57). In vertebrates, Pax7 has well-documented pioneer activity: it can engage nucleosomal DNA and deploy de novo enhancer repertoires, but efficient chromatin opening often requires a partner (for example Tpit, in the pituitary) (58–60). Pioneering and chromatin-opening functions of Pax7/Pax3 have also been documented in the myogenic lineages (61–63). In an oncogenic context, PAX3–FOXO1 fusion TF reprogrammes chromatin to install myogenic super-enhancers (64).

Pioneer binding does not guarantee activation. Across systems, pioneers can either seed opening (via co-activators and remodellers) or contribute to repression (via corepressors/HDACs), with the outcome set by partner availability and local chromatin context (55). For Pax7 specifically, binding and opening are separable: Pax7 can recognize heterochromatin on its own, but robust accessibility requires Tpit, illustrating why a pioneer may open some targets yet appear to close or dampen others when the right partners are missing (59). This division of labour aligns with our finding that Gsb remodels chromatin bidirectionally upon misexpression. More broadly, vertebrate Pax3/7 studies emphasize co-factor dependence, mirroring NB5-6, where Gsb-generated accessibility is insufficient for Hb recruitment.

En as a cofactor-dependent effector at pre-accessible enhancers

En in NB7-4 provides a counterpoint to Gsb. It binds almost exclusively to pre-accessible loci, with no detectable association with less accessible chromatin. This is in keeping with the wider literature: to our knowledge, neither *Drosophila* En nor its vertebrate homologues (EN1/EN2) have been shown to bind nucleosome-occluded DNA and initiate chromatin opening — the operational definition of pioneer activity (55). Thus, En lacks pioneer activity and likely acts as a recruiter or stabilizer with Hb at lineage-specific SoIs. En+Hb co-bound sites show the highest accessibility, consistent with cooperative enhancer activation. Because En does not open chromatin *de novo*, a different factor must prime NB7-4 enhancers - identifying this will be key to understanding lineage specification of NB7-4.

En can both activate and repress its target genes often via chromatin-modifying co-factors. As a repressor, En's eh1 motif recruits Groucho/TLE co-repressors, which in turn recruit HDAC/Rpd3, providing a route to local deacetylation and compaction (65–68). As an activator, En can partner with CBP/p300 (Nejire) at a regulatory element *in vivo*, linking En-dependent activation to histone acetylation (69). En can also act as a *bona fide* activator at specific targets (e.g., *polyhomeotic*) (69–71).

In summary, the best-supported chromatin mechanism for En is *cofactor-mediated modulation of already open enhancers* — activating via CBP/p300 and repressing via Gro/TLE — rather than pioneering access to closed chromatin, which fits our NB7-4 data.

The STF code for each NB

We invoke an “STF code” because of two lines of evidence: classic work shows that NB identity is combinatorial, set by orthogonal AP and DV patterning systems, and our data that reveals a division of labour among STFs (priming vs. effector roles), implying that no single STF suffices. Foundational mapping studies by Urbach and Technau explicitly framed NB identity as unique combinations of TFs (72). They linked 40–46 markers to individual brain NBs, effectively cataloguing NB-specific codes without yet resolving their mechanistic roles.

This suggests a practical definition: an NB's STF code is the minimal combination of STFs sufficient to confer lineage identity by delimiting the enhancer repertoire and specifying how the temporal cascade engages it—whether via STF-driven priming and TTF recruitment, direct co-occupancy on pre-accessible DNA, or, in some contexts, TTF-led chromatin remodelling with STFs selecting targets.

A corollary is testable: closely related NBs should be inter-converted by swapping the relevant code components. An extreme example of this was shown in the central brain where manipulation of a single TF, *otd*, in a pair of closely related NB lineages could completely change their identity rewiring both morphology and function (73). In general, however, the STF code predicts partial switches with manipulations of single factors (as seen for *gsb* or *vnd*) and more complete identity changes when multiple code members are comanipulated. Systematic code mapping across NBs — extending the Urbach and Technau atlases with perturbation and chromatin readouts — should reveal how modest changes in code composition generate the observed diversity of neuronal lineages.

Relation to vertebrate neural tube

Shared principles are seen in vertebrates. For example, spatial patterning has been exquisitely mapped in the spinal cord. Opposing morphogen gradients (ventral Shh, dorsal BMP/WNT) establish discrete progenitor domains along the dorsoventral axis, while Hox genes specify columnar and pool identities along A–P levels to diversify motor and interneuron classes (74). Temporal patterning, initially well defined in the developing cortex, has more recently has been generalized across vertebrate CNS regions, including the spinal cord. A shared temporal transcription factor code has been described that unfolds across progenitors to coordinate neuronal subtype production in time, complementing spatial patterning inputs (75).

These vertebrate studies raise a natural question: how are spatial and temporal cues integrated? A recent study suggests a distinct strategy for integrating spatial and temporal information. Zhang *et al.* report a global temporal chromatin program unfolding across progenitor domains that operates in parallel to spatial patterning: temporal regulators orchestrate the opening of regulatory elements over time, and spatial determinants act on the set of elements that become available. They describe this as a “chronotopic” integration strategy and it predicts SoI-like elements in vertebrates where temporal state broadly reconfigures accessibility and competence, onto which spatial TFs then act (76).

Materials and Methods

All fly stocks are listed in Table 1 [↗](#). All antibodies used are listed in Table 2 [↗](#).

Fly Stock Maintenance

Fly stocks were maintained on a standard cornmeal medium. The recipe consisted of the following per litre: 80g corn flour, 20g glucose, 40g sugar, 15g yeast extract, 4mL propionic acid, 5mL p-hydroxybenzoic acid methyl ester (in ethanol), and 5mL ortho-butyric acid. Flies were raised at 25°C under a 12h:12h light-dark cycle. The virginator and *hsGsb* stocks were grown at room temperature.

Generation of Dam:Gsb, Dam:En, and gsb-guideRNA fly lines

To generate the gsb-guideRNA line, *Drosophila* gsb-guideRNA sequences were cloned into the tandem guideRNA expression vector pCFD4 ((77)). The gRNA target sequences were amplified via PCR using primers, and the resulting PCR product was assembled into the BbsI-digested pCFD4 backbone using Gibson assembly.

TaDa and CATaDa Experiments

Verification of Dam:Gsb and Dam:En

To assess whole-embryo occupancy of Gsb and En, Dam fusion constructs were expressed using the ubiquitous *Da-Gal4* driver. 1000-1600 females of *w-;;Da-Gal4* were crossed to a similar number of males of *w-;;UAS-LT3-Dam:Gsb* for Gsb occupancy, *w-;;UAS-LT3-Dam:En* for En occupancy, and *w-;;UAS-LT3-DamOnly* for chromatin accessibility.

After 16–24 hours of mating, adults were transferred to embryo collection cages. Collection plates were prepared using 2% agar and 2% sucrose in 60–100 mm Petri dishes, and yeast paste was applied to stimulate egg laying. Embryos were collected every 4 hours (0–4 h AEL) and then aged for 12 hours (12–16 h AEL; Stages 15–16) at 25°C.

Following ageing, embryos were rinsed with distilled water and dechorionated in bleach for 2 minutes, or until complete removal of the chorion was visually confirmed. Embryos were then washed thoroughly with distilled water, dried, weighed, and stored in 1.5mL tubes at –20°C. For each biological replicate, a minimum of 30 mg of embryos was collected.

Determination of NB-specific STF occupancy and chromatin accessibility

To determine Gsb occupancy and chromatin accessibility in NB5-6, we crossed 4000 males of *w-;;Lbe-k-Gal4* to 4000 females of *w-;;UAS-LT3-Dam:Gsb* or *w-;;UAS-LT3-DamOnly*. Similarly, to determine En binding and chromatin accessibility in NB7-4, 4000 males of *w-;;19B03[AD];18F07[DBD]* were crossed to 4000 females of *w-;;UAS-LT3-Dam:En* or *w-;;UAS-LT3-DamOnly*. Embryos were collected for 2h (0-2h AEL) and then aged for 6h (6h-8h AEL, Stage 11-12) at 25°C. We collected 4 replicates of at least 30mg of embryos each for NB5-6-specific Dam:Gsb and DamOnly, and 3 replicates each of NB7-4-specific Dam:En and DamOnly.

Reagent type (species) or resource	Name	Source or Reference	Identifiers	Additional information
Strain, strain background (<i>Drosophila melanogaster</i>)	UAS-LT3-Dam:Gsb	This paper	NA	UAS drives mCherry from 1st cistron and Dam fused to Gsb (Dam:Gsb) from 2nd cistron
Strain, strain background (<i>Drosophila melanogaster</i>)	UAS-LT3-Dam:En	This paper	NA	UAS drives UAS drives mCherry from 1st cistron and Dam fused to En (Dam:En) from 2nd cistron
Strain, strain background (<i>Drosophila melanogaster</i>)	UAS-LT3-DamOnly	A. Brand	NA	UAS drives mCherry from 1st cistron and Dam from 2nd cistron
Strain, strain background (<i>Drosophila melanogaster</i>)	UAS-LT3-DamHb	Sen <i>et al.</i> 2019 (41)	NA	UAS drives mCherry from 1st cistron and Dam fused to Hb (Dam:Hb) from 2nd cistron
Strain, strain background (<i>Drosophila melanogaster</i>)	hsGsb;;	Doe Lab	NA	Ubiquitously misexpresses Gsb upon heat shock
Strain, strain background (<i>Drosophila melanogaster</i>)	Da-Gal4	BDSC	55850	Ubiquitous expression in the embryo
Strain, strain background (<i>Drosophila melanogaster</i>)	Lbe-k-Gal4	S. Thor	NA	Expression in NB5-6 lineage from stage 9; salivary gland at stage 17
Strain, strain background (<i>Drosophila melanogaster</i>)	R19B03[AD]; R18F07[DBD]	G. Rubin	NA	Expression in NB7-4 lineage from stage 9
Strain, strain background (<i>Drosophila melanogaster</i>)	R59E09-Gal4	G. Rubin	NA	Expression in NB3-5 lineage from stage 9
Strain, strain background (<i>Drosophila melanogaster</i>)	Vasa-Cas9 (y[1] MRFP[3xP3.PB] GFP[E.3xP3]=vasa-cas9ZH-2A w[1118]/FM7c)	Bangalore Fly Facility	NA	Cas9 expression in germline cells driven by the <i>vasa</i> promoter
Strain, strain background (<i>Drosophila melanogaster</i>)	gsb-gRNA K/O (y1v1; pssl.pcFD4 attp40/CyO)	Bhagyashree Kaduskar, This paper	NA	Produces indels at the gsb locus
Strain, strain background (<i>Drosophila melanogaster</i>)	w-/hsHid(Y);; MKRS/TM2	BDSC	24640	Virginator stock; heat-shocks given at the L3 to early pupal stage at 37°C for 30 minutes on two consecutive days, after which they were moved to 25°C.

Table 1. Resource Table: List of fly lines used.

Reagent type (species) or resource	Name	Source or Reference	Identifiers	Additional information
Primary Antibody	Chicken anti-GFP (polyclonal)	Abcam	ab13970	1:1000
Primary Antibody	Rabbit anti-Apterous	Gift from Juan Botas	NA	1:1000
Primary Antibody	Mouse anti-BP102	DSHB	528099	1:200
Primary Antibody	Rat anti-CCAP	Gift from Fernando J. Díaz-Benjumea	NA	1:100
Primary Antibody	Rat anti-CD8	Abcam	22378	1:200
Primary Antibody	Rat anti-Ems	Gift from Uwe Waldorf	NA	1:500
Primary Antibody	Mouse anti-En	DSHB	528224	1:200
Primary Antibody	Mouse anti-Eya	DSHB	528232	1:200
Primary Antibody	Mouse anti-fasII	DSHB	528235	1:50
Secondary Antibody	Goat anti-Chicken Alexa Fluor 488	Invitrogen	A32931TR	1:400
Secondary Antibody	Goat anti-Mouse Alexa Fluor 568	Invitrogen	A-11004	1:400
Secondary Antibody	Goat anti-Mouse Alexa Fluor 647	Invitrogen	A-21235	1:400
Secondary Antibody	Goat anti-Rabbit Alexa Fluor 568	Invitrogen	A-11011	1:400
Secondary Antibody	Goat anti-Rabbit Alexa Fluor 647	Invitrogen	A-21245	1:400
Secondary Antibody	Goat anti-Rat Alexa Fluor 568	Invitrogen	A-11077	1:400
Secondary Antibody	Goat anti-Rat Alexa Fluor 647	Invitrogen	A-21247	1:400

Table 2. Resource Table-Antibodies Used

Determination of NB7-4-specific Hb occupancy and chromatin accessibility, upon broad Gsb misexpression

To assay Hb occupancy and chromatin accessibility in NB7-4 under conditions of broad Gsb misexpression, we crossed 5000 females each of *hsGsb;;UAS-LT3-DamOnly* and *hsGsb;;UAS-LT3-Dam:Hb* to males of *w⁻;19B03[AD];18F07[DBD]*. Embryos were collected every hour (0-1h AEL), incubated at 25°C for 3h (3-4h AEL, Stage 7-8) and then heat shocked at 33°C just before NB delamination for 4h (7-8h AEL, Stage 11-12). Sufficient embryos were collected for 3 replicates per condition.

TaDa/CATaDa Experimental Pipeline

TaDa and CATaDa replicates were processed according to (78). Briefly, embryos were homogenized using motorized pestles, and genomic DNA was extracted using the QIAamp DNA Micro Kit (Qiagen, 56304) in a final volume of 50 µL. To minimise genomic DNA shearing, wide-bore tips were used for all pipetting steps. DNA quality was assessed by loading 3 µL onto a 0.8% agarose gel. Genomic DNA was then digested with DpnI (NEB, R0176S) to cut at sites methylated by Dam or Dam:TF proteins. This was followed by DamID adaptor ligation and DpnII digestion (NEB, R0543S) to fragment unmethylated DNA. Finally, the adaptor-ligated DNA fragments were amplified by PCR (21 cycles) using MyTaq HS DNA Polymerase kit (Bioline, BIO-21112). Sequencing libraries were prepared at the Next Generation Genomics Facility (NGGF) at Bangalore Life Science Cluster (BLiSC), using the NEBNext® Ultra™ II DNA Library Prep with Sample Purification Beads (E7103L), and sequenced on the Illumina HiSeq2500 (50 base pairs single-end, 20–60 million single end reads per sample). For TaDa/CATaDa experiments involving Gsb misexpression, samples were sequenced on the NovaSeq 6000 platform (50 base pairs, 25-30 million paired-end reads per sample).

Data processing and analysis

Alignment and Filtering

Reads were aligned to the *Drosophila* genome (dm6) using Bowtie2 (79). The .sam files thus obtained were sorted and converted to .bam files using the SAMtools (80) and NGSUtils (81) suites. We removed reads with poor mapping quality, as well as those that aligned with blacklisted regions in dm6 or the mitochondrial genome. The resultant .bam files were used for all downstream analyses.

Pearson correlation plots were generated using the *multiBam-Summary* and *plotCorrelation* tools from the deepTools suite (82).

Peak Calling and Differential Analysis

We used MACS2 (v2.2.6) (83) to call narrow peaks on Dam:TF or DamOnly files, without controls. We used the *–nolambda* parameter to disable local background correction, instead calling peaks relative to a global background.

Differential analyses were performed on these peaks using DiffBind (84, 85).

(i) Determining Sites of Integration (SoIs)

SoIs were defined using Hb-TaDa data from NB5-6 and NB7-4 generated in (41). Differential analysis was performed using DiffBind, and peaks were considered differential if they met the criteria of $FDR \leq 0.01$ and ≥ 2 -fold change between NB5-6 and NB7-4. Shared Hb-bound sites were identified using *bedtools intersect* with a minimum overlap of $\geq 50\%$.

(ii) Determining genome-wide differentially accessible sites in NB5-6 and NB7-4

Genome-wide accessible and less accessible loci in NB5-6 and NB7-4 were identified by performing a differential analysis using DiffBind on Dam-bound sites in the two NBs. Peaks were considered differential if they met the criteria of $FDR \leq 0.01$ and ≥ 2 -fold change. To identify loci accessible in both NB5-6 and NB7-4, we used *bedtools intersect* with a minimum overlap of $\geq 50\%$.

(iii) Determining genome-wide STF-bound and -unbound sites

STF-bound and -unbound sites in NB5-6 and NB7-4 were determined by performing a differential analysis using DiffBind between Gsb-bound sites in NB5-6 and En-bound sites in NB7-4. Peaks were considered differential if they satisfied the thresholds of $FDR \leq 0.01$ and ≥ 2 -fold change. Sites bound by STFs in both NBs were defined using *bedtools intersect* with a minimum overlap of $\geq 50\%$.

(iv) Determining genome-wide loci that are STF-only, TTF-only and STF + TTF co-bound

For this analysis, the top 5000 most confident peaks were selected for each TF—Gsb in NB5-6, En in NB7-4, and Hb in both NB5-6 and NB7-4. For each TF, these 5000 peaks included their known genomic targets. *Bedtools intersect* with the -v parameter was used to identify STF-only and TTF-only sites, while co-bound (STF + TTF) sites were defined using *bedtools intersect* with $\geq 90\%$ overlap.

(v) Determining genome-wide changes in accessibility in NB7-4 upon Gsb misexpression

We used *hsGsb* to induce embryo-wide Gsb misexpression and assayed changes in chromatin accessibility and Hb binding in NB7-4 under these conditions. We performed a differential analysis between Dam binding in NB7-4 with and without ectopic Gsb expression. Sites were considered differential at a threshold of $FDR \leq 0.01$ and ≥ 2 -fold change.

Signal Correlation

We obtained signal files by using *bam-Coverage* from deepTools to convert all .bam files to .bigWig files. All signal files were normalized to $1 \times$ genome coverage (reads per genomic content, RPGC), thereby correcting for differences in sequencing depth between samples. All signal files were visualized using IGV Genome Browser (86, 87). To generate signal correlation plots across different peaksets, we used *computeMatrix* (from deepTools (82)) to obtain matrices of per-bin signal values for each replicate along a 3kb region upstream and downstream from the peak centre (bin size = 10). These matrices were imported into R, where mean signals across replicates were calculated and plotted with 95% confidence intervals. We similarly generated heat maps showing mean signals across replicates to visualize binding signal for each site.

Gsb knockdown and misexpression experiments

For knockdown experiments, crosses were set up with *Vasa-Cas9*; and *y[1]v[1]; gsb-gRNA*. Appropriately staged embryos were collected and stained with *Rabbit anti-Ap*, *Mouse anti-Eya* and *Rat anti-Ems*. Stained embryos were imaged. Marker-positive cells per hemisegment were counted. The data was plotted using R.

For misexpression experiments, crosses were set up with *hsGsb*; and *w-;Lbe-k-Gal4,UAS-mCD8-GFP* (to visualize NB5-6 lineage) or *w-;R59E09,UAS-myrGFP* (to visualize NB3-5 lineage). Embryos were incubated in 18°C for 6.5h and then given a heat shock at 37°C for 30min. Subsequently, they were incubated at 23°C for 21h to stage 16 and then stained with *Chicken anti-GFP*, *Mouse anti-Eya*, *Rabbit anti-Ap*, and *Rat anti-Ems*. Marker-positive cells per hemisegment were counted. The data was plotted using R.

In all cases, box and whisker plots were generated. Statistical significance was assessed using two-tailed Mann–Whitney U tests. Significance levels are indicated as: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

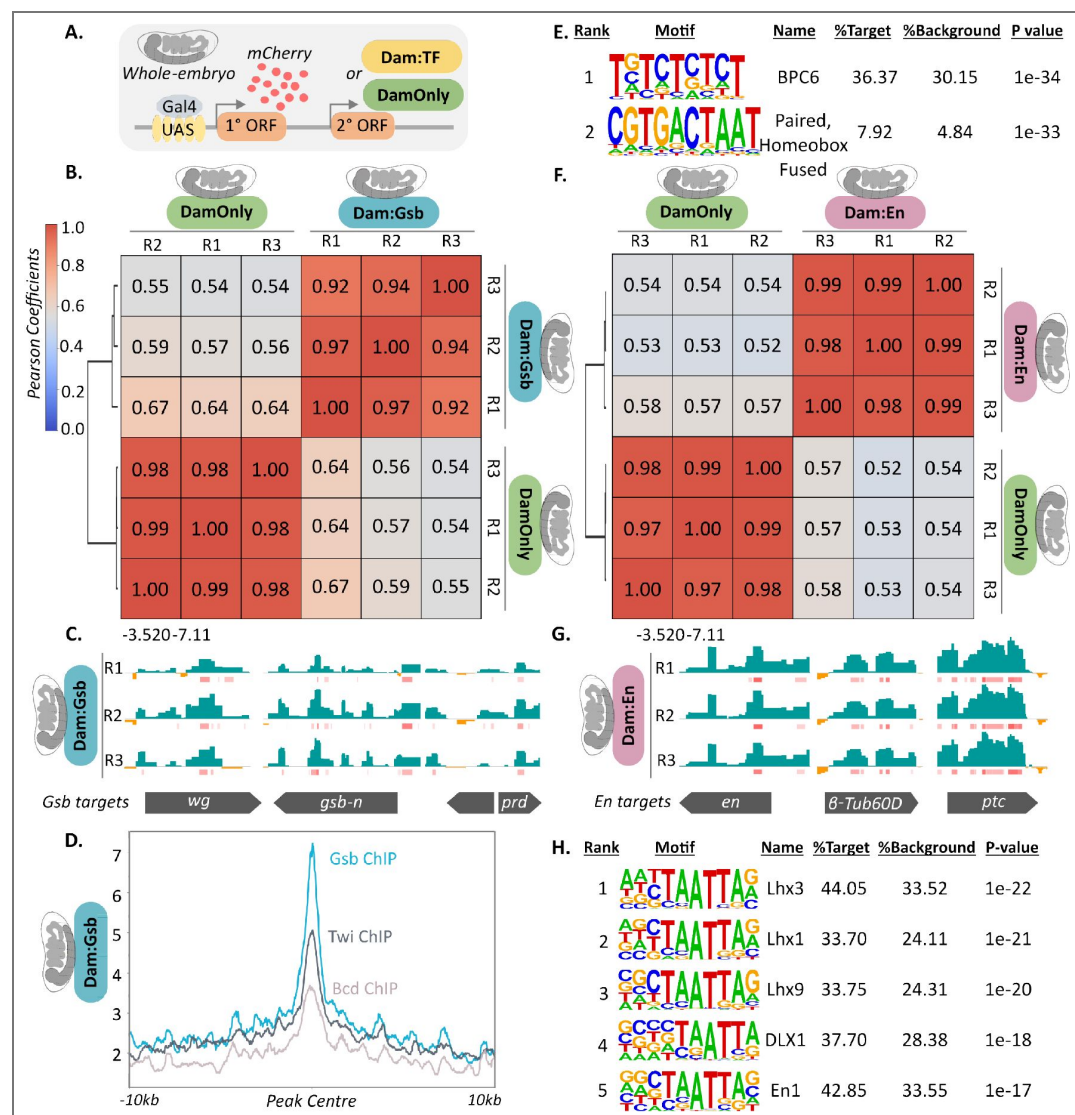
Immunohistochemistry and imaging

Appropriately staged embryos were washed with distilled water and dechorionated in bleach for 2–3 minutes. After thorough rinsing with distilled water, embryos were fixed by nutating in a 1:1 mixture of 4% PFA and heptane for 20–25 minutes. Vitelline membranes were removed by vigorous shaking for 1 minute in a 1:1 mixture of heptane and methanol. Embryos were then washed in 0.5% PTX (0.5% Triton X-100 in 1X PBS), incubated overnight at 4°C with primary antibodies, washed again in 0.5% PTX, and incubated overnight at 4°C with secondary antibodies. Following additional washes in 0.5% PTX, embryos were cleared in 90% glycerol with Vectashield.

Imaging was performed at the Central Imaging and Flow Cytometry Facility (CIFF), Bangalore Life Science Cluster (BLISC), using an Olympus FV3000 system with a 60X oil-immersion objective. Images were acquired at 512×512 or 1024×1024 resolution, viewed and manually analyzed in FIJI (ImageJ) (88), and are presented as maximum-intensity projections of z-stacks.

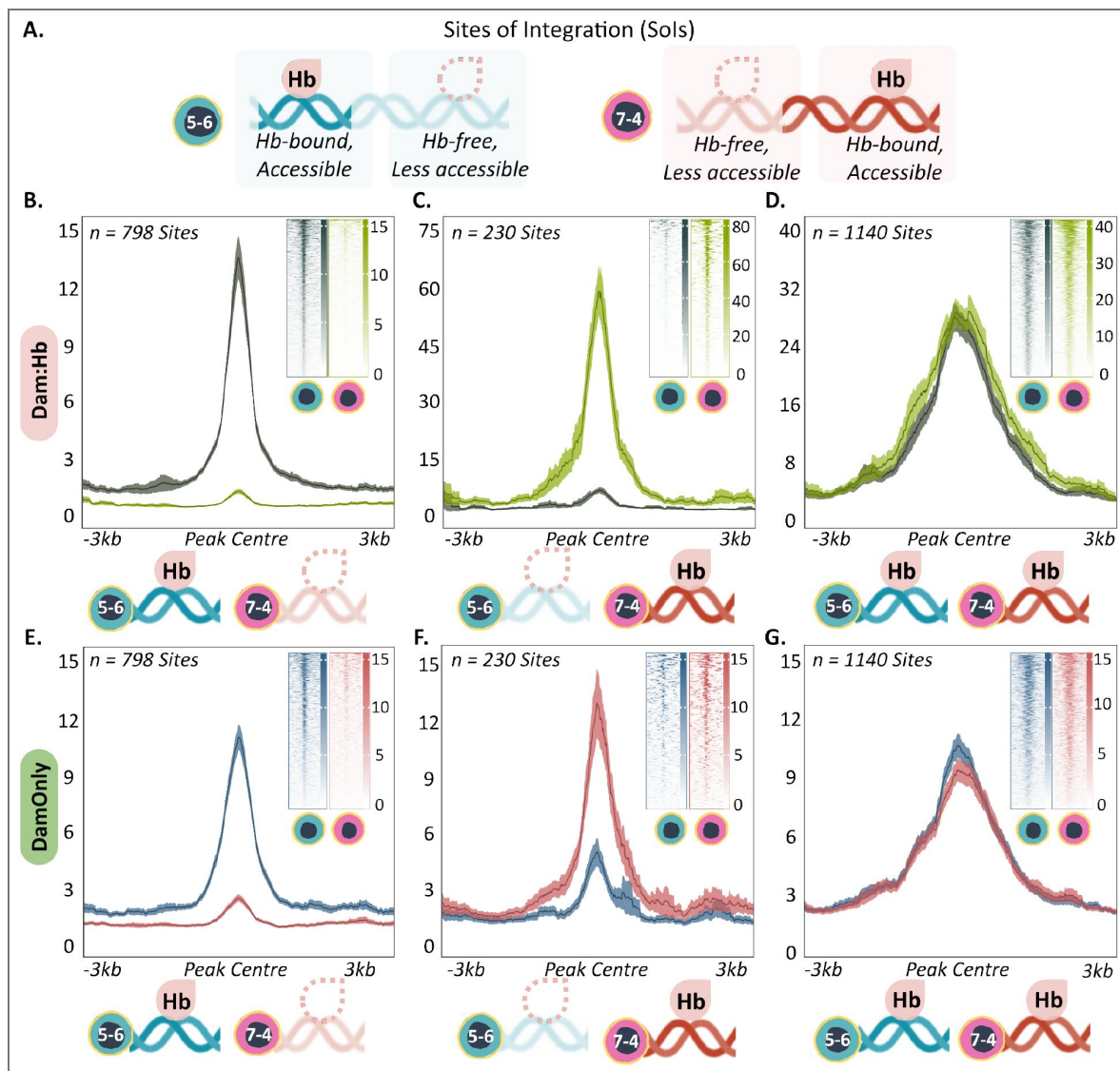
All figures have been assembled using Inkscape 1.4.2 (f4327f4, 2025-05-13).

Supplementary figures



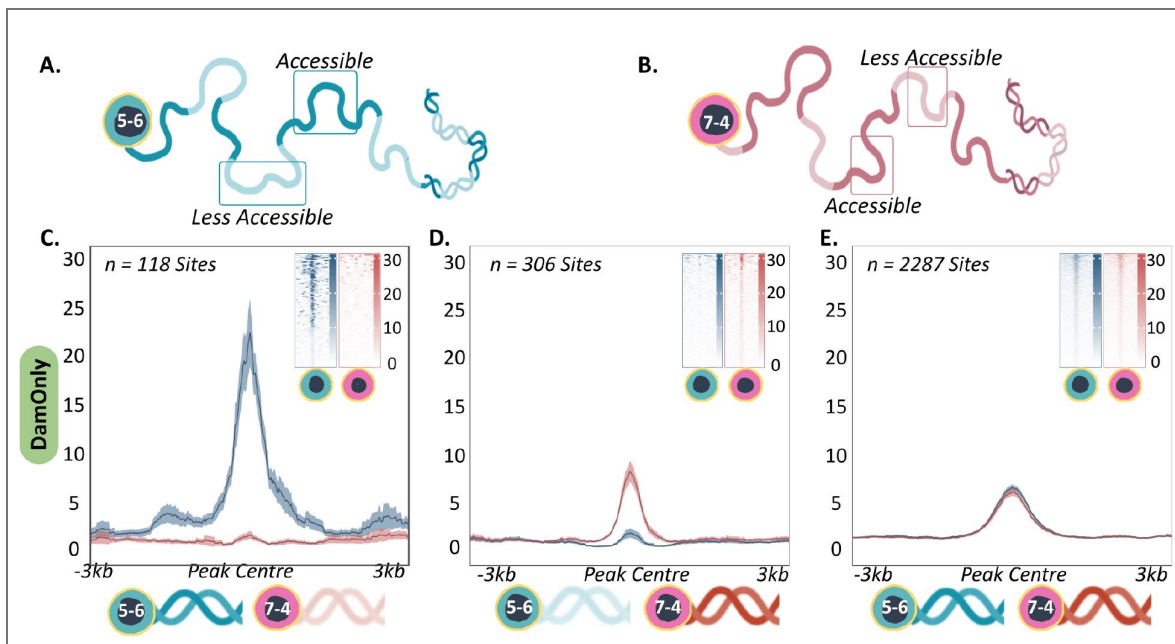
Supplementary Fig. S1. Verification of TaDa Tools. **A. Design of TaDa and CATaDa tools:** The ubiquitous *Da-Gal4* driver expresses mCherry from the primary ORF and either DamOnly (green) or Dam:TF (yellow) from the secondary ORF. High mCherry expression reports driver specificity, while translation of Dam fusions from the secondary ORF keeps Dam levels low, minimizing toxicity and preserving binding specificity. **B. Whole-embryo DamOnly and Dam:Gsb bind reproducibly:** Heatmap shows pairwise Pearson correlations between DamOnly and Dam:Gsb replicates. Hierarchical clustering separates DamOnly and Dam:Gsb samples, with stronger correlations within than between conditions. **C. Binding at known *Gsb* targets:** Genome browser snapshots of log2 ratio files (Dam:Gsb/DamOnly) are shown (3 replicates), Dam:Gsb in blue positive tracks, DamOnly in orange negative tracks. Replicates show occupancy at *wg* (46), *gsb-n* (47), *prd* (48). Data range -3.520-7.11. Pink bars indicate peaks, colour intensity indicates peak score. **D. TaDa-Gsb correlates with published *Gsb* ChIP-seq data:** Dam:Gsb signals are enriched at *Gsb*-ChIP peaks but not at *Twi*-ChIP or *Bcd*-ChIP sites (43). **E. Motif enrichment at Dam:Gsb sites:** *de novo* motif analysis at Dam:Gsb peaks reveals a fused Paired-Homeobox motif. Together (B-

E) show that Dam:Gsb reproducibly and specifically reports Gsb occupancy in the chromatin. **F. Whole-embryo DamOnly and Dam:En bind reproducibly:** Heatmap shows pairwise-correlation between replicates of DamOnly and Dam:En. Hierarchical clustering shows that DamOnly and Dam:En replicates cluster separately, with stronger correlations within than between conditions. **G. Binding at known En targets:** Genome browser snapshots of log2 ratio files (Dam:En/DamOnly) are shown (3 replicates), Dam:En in blue positive tracks, DamOnly in orange negative tracks. Replicates show occupancy at *en* (49), *beta-tub60D* (50) and *ptc* (49). Data range -3.520-7.11. Pink bars indicate peaks, colour intensity indicates peak score. **H. Motif enrichment at Dam:En sites:** See entry at Rank 5, p-value 1e-17. Together, **(F-H)** show that Dam:En reproducibly and specifically reports En binding in chromatin. Genotypes for embryo-wide Gsb occupancy: *w-;; Da-Gal4/UAS-LT3-Dam:Gsb*. For embryo-wide En occupancy: *w-;; Da-Gal4/UAS-LT3-Dam:En*. For embryo-wide chromatin accessibility: *w-;; Da-Gal4/UAS-LT3-DamOnly*.



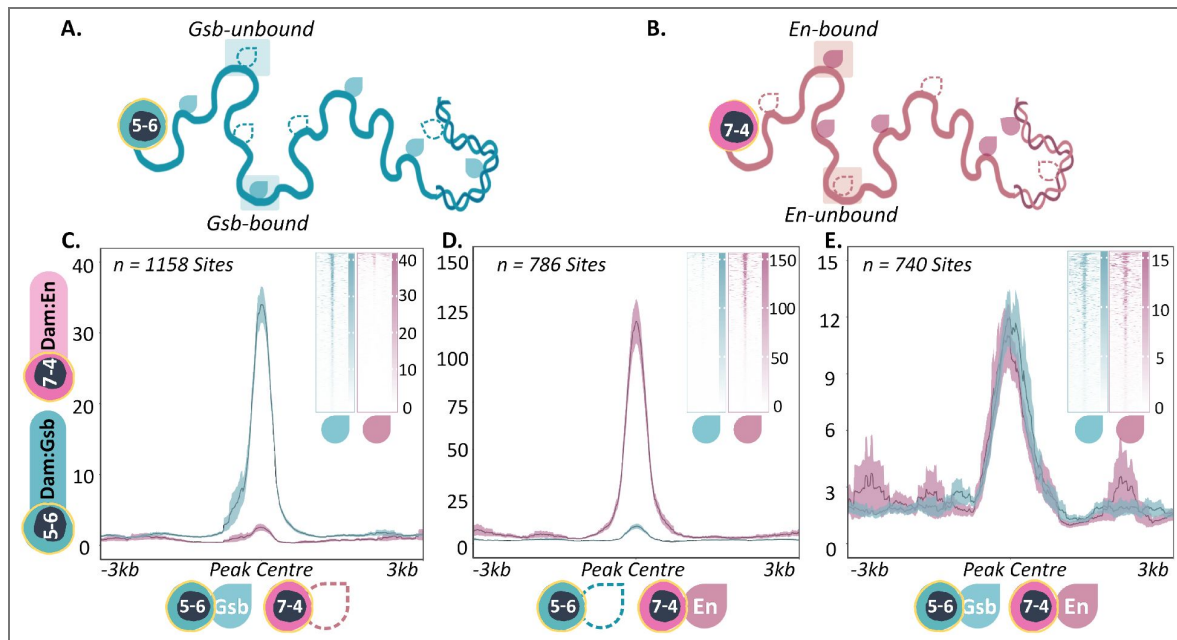
Supplementary Fig. S2. NB-specific sites of integration (SoIs) are differentially accessible (41).

A. Schematic representing NB5-6-SoIs (798 sites), defined as loci that are Hb-bound (light pink drop) and accessible in NB5-6 (dark blue DNA strand), but remain Hb-free (hollow drop, dotted light-pink outline) and less accessible in NB7-4 (light red DNA strand). Conversely, NB7-4-SoIs (230 sites) are Hb-bound (light pink drop) and accessible in NB7-4 (dark red DNA strand), while the corresponding loci in NB5-6 are Hb-free (hollow drop, dotted light-pink outline) and less accessible (light blue DNA strand) (FDR ≤ 0.01 and FC ≥ 2). **B-D.** At NB5-6-SoIs, Hb signal from NB5-6 (dark grey) is higher than that from NB7-4 (light green) (**B**). Conversely, at NB7-4-SoIs, Hb signal from NB7-4 is higher than that from NB5-6 (**C**). At shared Hb-bound loci (1140 sites, Hb-bound in both NBs with $\geq 50\%$ overlap), Hb binding signal from both NBs is similar. 3 Hb replicates each from NB5-6 and NB7-4 have been averaged. Solid lines indicate mean; coloured ribbons indicate 95% confidence intervals. Insets show heatmaps of Hb signal from NB5-6 and NB7-4 at the corresponding loci, averaged across replicates. **E-G.** At NB5-6-SoIs, Dam signal shows chromatin is accessible in NB5-6 (blue) but less accessible in NB7-4 (red) (**E**). At NB7-4-SoIs, Dam signal shows chromatin is more accessible in NB7-4 than in NB5-6 (**F**). At shared Hb-bound sites, chromatin is similarly accessible in both NBs (**G**). Together (**B-G**) validates computationally determined SoIs. 4 replicates of Dam from NB5-6 have been averaged, as have 3 replicates of Dam from NB7-4. Solid lines indicate mean, coloured ribbon indicates 95% confidence interval. Insets show heatmaps of Dam from NB5-6 and NB7-4 at the corresponding loci, averaged across replicates. Genotypes for NB5-6-Hb occupancy: *w-; Lbe-k-Gal4/+; +/UAS-Dam:Hb*. NB7-4-Hb occupancy: *w-; 19B03[AD]/+; 18F07[DBD]/UAS-Dam:Hb* (41). NB5-6-chromatin accessibility: *w-; Lbe-k-Gal4/+; +/UAS-DamOnly*. NB7-4-chromatin accessibility: *w-; 19B03[AD]/+; 18F07[DBD]/UAS-DamOnly*.



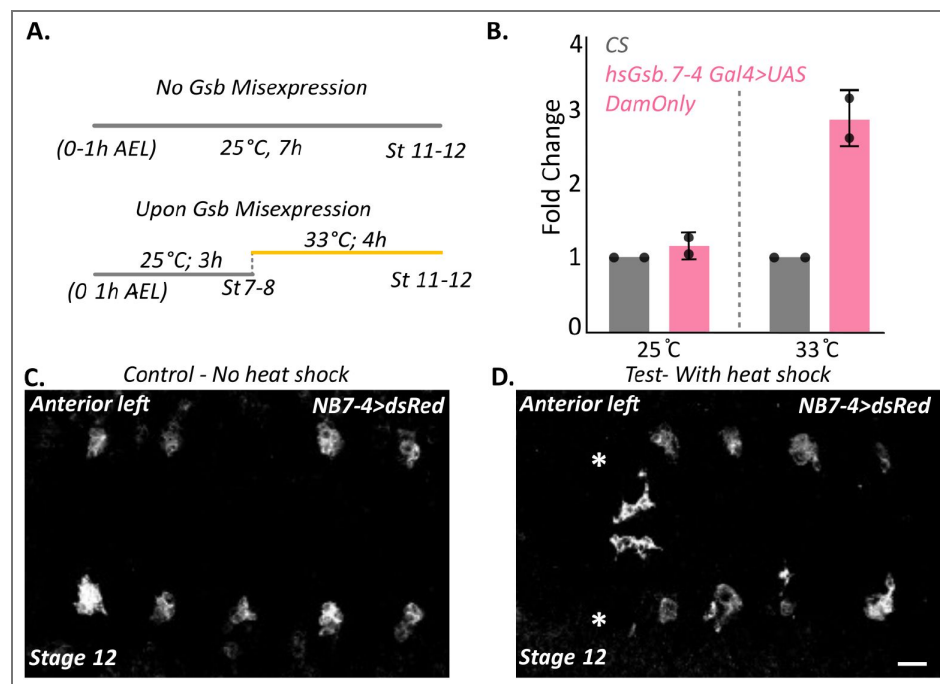
Supplementary Fig. S3. Validation of NB-specific accessible sites.

A. Schematic showing uniquely accessible (dark blue) and less accessible (light blue) loci in NB5-6. **B.** Schematic showing uniquely accessible (dark red) and less accessible (light red) loci in NB7-4. **C-E. Dam signal at differentially accessible chromatin:** At sites uniquely accessible in NB5-6 but less accessible in NB7-4 (118 sites), Dam signal from NB5-6 (blue) is higher than that from NB7-4 (red) (**C**). At sites uniquely accessible in NB7-4 but less accessible in NB5-6 (306 sites) Dam signal from NB7-4 is higher than from NB5-6 (**D**). At sites accessible in both NBs (2287 sites, Dam-bound in both NBs with $\geq 50\%$ overlap), Dam signal from both NBs are similar. Together, (**C-E**) validate computationally determined NB-specific accessible sites. 4 replicates of Dam from NB5-6 have been averaged, as have 3 replicates of Dam from NB7-4. Solid lines indicates mean, coloured ribbons indicate 95% confidence interval. Insets show heatmaps of Dam signal from NB5-6 and NB7-4 at the corresponding loci, averaged across replicates. Genotypes for NB5-6-chromatin accessibility: *w-; Lbe-k-Gal4/+; +/UAS-DamOnly*. NB7-4-chromatin accessibility: *w-; 19B03[AD]/+; 18F07[DBD]/UAS-DamOnly*.



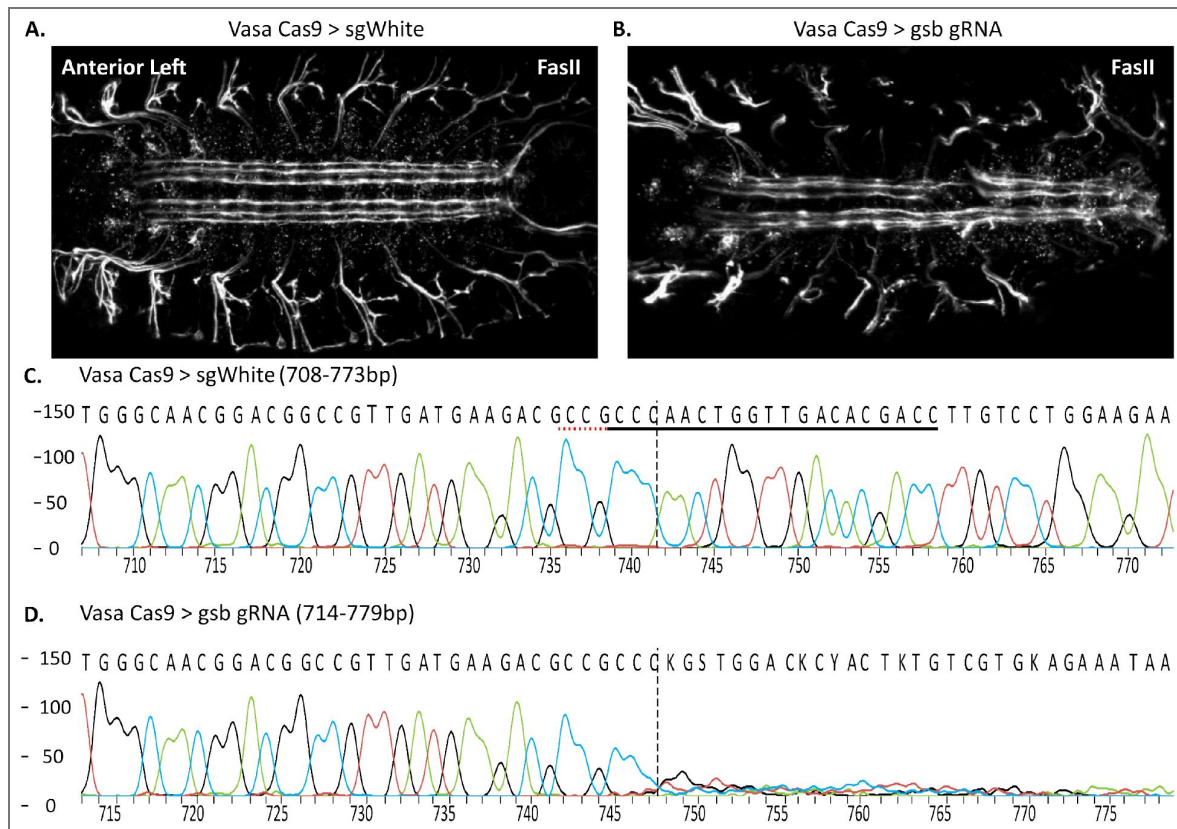
Supplementary Fig. S4. Validation of computationally determined STF-bound and -unbound sites.

A. Schematic representing Gsb-bound (blue drops) and -unbound sites (hollow drops, dotted blue outline). **B.** Schematic representing En-bound (pink drops) and -unbound sites (hollow drops, dotted pink outline). Sites defined as 'Gsb-bound' in NB5-6 are 'En-unbound' in NB7-4 (1158 sites). Conversely, sites that are 'En-bound' in NB7-4 are 'Gsb-unbound' in NB5-6 (786 sites) ($FDR \leq 0.01$ and $FC \geq 2$). **C-E. STF signal at STF-bound/unbound sites:** At Gsb-bound/En-unbound sites, Gsb signal from NB5-6 (blue) is higher than En signal from NB7-4 (pink) (**C**). At Gsb-unbound/En-bound sites, En signal from NB7-4 is higher than Gsb signal from NB5-6 (**D**). At sites STF-bound in both NBs (740 sites, NB5-6-Gsb-bound and NB7-4-En-bound peaks, with $\geq 50\%$ overlap), Gsb signal from NB5-6 is similar to En signal from NB7-4. Together (**C-E**) validate computationally determined STF-bound/unbound sites. 4 replicates of Gsb from NB5-6 have been averaged, as have 3 replicates of En from NB7-4. Solid lines indicates mean, coloured ribbons indicate 95% confidence interval. Insets show heatmaps of Gsb from NB5-6 and En from NB7-4 at the corresponding loci, averaged across replicates. Genotypes for NB5-6-Gsb occupancy: *w-; Lbe-k-Gal4/+*; *+UAS-Dam:Gsb*. NB7-4-En occupancy: *w-; 19B03[AD]/+; 18F07[DBD]/UAS-Dam:En*.



Supplementary Fig. S5. Validation of heat shock regime for Gsb misexpression.

A. Schematic showing heat shock protocol for sample collection. Embryos were collected (0h-1h AEL) and incubated at 25°C (control, no Gsb misexpression) or moved to 33°C (test, with Gsb misexpression) pre-delamination (Stage 7-8). **B. Increased Gsb expression upon heat shock:** qPCR results show increase in Gsb transcripts relative to wild-type CS embryos, under the heat shock protocol outlined in (A). **C-D. Continued Gal4 expression upon Gsb misexpression:** Crosses were set up to obtain embryos of the genotype *hsGsb;19B03[AD]/UAS-mCD8-dsRed;18F07[DBD]*, with or without heatshock. (C-D) show that this NB7-4-specific Gal4 expression stays on under the heat shock protocol outlined in (A). White asterisks mark missing NB7-4 lineages. Scale bar 10μ.



Supplementary Fig. S6. Validation of *gsb*-guideRNA.

A. In control embryos (Vasa Cas9 > sgWhite), Vasa-Cas9 is directed to the *white* gene locus by the sgWhite guideRNA. FasII staining in these embryos shows intact axonal tracts. **B.** In test embryos (Vasa Cas9 > gsb gRNA), axonal tract defect is seen, consistent with loss of Gsb in embryos. **C-D.** Sanger sequencing chromatograms confirm Cas9-mediated editing at the *gsb* locus. Sequencing chromatograms of embryos expressing Vasa-Cas9 and sgWhite control (C) or gsb-guideRNA (Edited sample, (D)) were analyzed using the Synthego ICE (Inference of CRISPR Edits) online tool. The predicted Cas9 cut site (dashed black line) corresponds to position 747 bp. Mixed peaks downstream of the cut site in the edited sample indicate indel formation at the *gsb* target locus, consistent with efficient CRISPR editing. The control trace shows clean base calls with no evidence of indels. ICE predicts 75% editing efficiency. Genotypes for control embryos: *Vasa Cas9*^{+/+}; *sgWhite*^{+/+}. For test embryos: *Vasa Cas9*^{y[1]v[1]}; *gsb-guideRNA*^{+/+}.

Data availability

This study has generated DNA sequence data. The raw and processed files have been deposited to NCBI and already made public. The BioProject accession number is PRJNA1372925.

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Additional information

Author contributions

AB conceptualised the project, performed and analysed all experiments, wrote and reviewed the manuscript. HR conceptualised, performed, and analysed experiments related to the functional analysis of Gsb and reviewed the manuscript. SQS conceptualised and supervised the project and wrote and reviewed the manuscript.

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Additional files

Supplementary Table S1. [Summary of TaDa and CATaDa sequencing quality metrics.](#)

Supplementary Table S2. [Genomic coordinates of Sites of Integration \(SoIs\).](#)

Supplementary Table S3. [Genomic coordinates of differentially accessible sites in NB5-6 and NB7-4.](#)

Supplementary Table S4. [Genomic coordinates of STF-bound and -unbound sites.](#)

Supplementary Table S5. [Genomic coordinates of sites that are bound by STF only, Hb only, and co-bound by both STF and Hb in NB5-6 and NB7-4.](#)

Supplementary Table S6. [Genomic coordinates of sites that change accessibility upon Gsb misexpression.](#)

Supplementary Table S7. [Raw quantification of marker-positive cells per hemisegment \(Ap, Ems, Eya, GFP-reporters\) across all experimental conditions. This sheet contains the complete dataset used for all quantifications shown in Figures 7–8.](#)

References

1. **Bakshi Asif**, Ben El Kadhi Khaled, Desplan Claude (2025) Decoding neuronal diversity: Mechanisms governing neural cell fate in *Drosophila*. *Current Opinion in Neurobiology* **93**:103061 <https://doi.org/10.1016/j.conb.2025.103061>
2. **Rossi Anthony M.**, Jafari Shadi, Desplan Claude (2021) Integrated Patterning Programs During *Drosophila* Development Generate the Diversity of Neurons and Control Their Mature Properties. *Annual Review of Neuroscience* **44**:153-172 <https://doi.org/10.1146/annurev-neuro-102120-014813>
3. **Sen Sonia Q.** (2023) Generating neural diversity through spatial and temporal patterning. *Seminars in Cell & Developmental Biology* **142**:54-66 <https://doi.org/10.1016/j.semcdb.2022.06.002>
4. **Schmid Aloisia**, Chiba Akira, Doe Chris Q. (1999) Clonal analysis of *Drosophila* embryonic neuroblasts: Neural cell types, axon projections and muscle targets. *Development* **126**:4653-4689 <https://doi.org/10.1242/dev.126.21.4653>
5. **Schmidt Hartmut**, Rickert Christof, Bossing Torsten, Vef Olaf, Urban Joachim, Technau Gerhard M. (1997) The embryonic central nervous system lineages of *Drosophila melanogaster*. *Developmental Biology* **189**:186-204 <https://doi.org/10.1006/dbio.1997.8660>
6. **Lacin Haluk**, Truman James W (2016) Lineage mapping identifies molecular and architectural similarities between the larval and adult *Drosophila* central nervous system. *eLife* **5**:e13399 <https://doi.org/10.7554/eLife.13399>
7. **Bossing Torsten**, Udolph Gerald, Doe Chris Q., Technau Gerhard M. (1996) The embryonic central nervous system lineages of *Drosophila melanogaster*: I. Neuroblast lineages derived from the ventral half of the neuroectoderm. *Developmental Biology* **179**:41-64 <https://doi.org/10.1006/dbio.1996.0240>
8. **Landgraf Matthias**, Bossing Torsten, Technau Gerd M., Bate Michael (1997) The Origin, Location, and Projections of the Embryonic Abdominal Motorneurons of *Drosophila*. *Journal of Neuroscience* **17**:9642-9655 <https://doi.org/10.1523/JNEUROSCI.17-24-09642.1997>
9. **Skeath J. B.**, Panganiban G., Selegue J., Carroll S. B. (1992) Gene regulation in two dimensions: The proneural achaete and scute genes are controlled by combinations of axis-patterning genes through a common intergenic control region. *Genes & Development* **6**:2606-2619 <https://doi.org/10.1101/gad.6.12b.2606>
10. **Skeath James B.**, Zhang Yu, Holmgren Robert, Carroll Sean B., Doe Chris Q. (1995) Specification of neuroblast identity in the *Drosophila* embryonic central nervous system by gooseberry-distal. *Nature* **376**:427-430 <https://doi.org/10.1038/376427a0>
11. **Chu-LaGraff Quynh**, Doe Chris Q. (1993) Neuroblast Specification and Formation Regulated by wingless in the *Drosophila* CNS. *Science* **261**:1594-1597 <https://doi.org/10.1126/science.8372355>
12. **Lundell Martha J.**, Chu-LaGraff Quynh, Doe Chris Q., Hirsh Jay (1996) The engrailed and huckebein genes are essential for development of serotonin neurons in the *Drosophila* CNS. *Molecular and Cellular Neuroscience* **7**:46-61 <https://doi.org/10.1006/mcne.1996.0004>
13. **Deshpande Nirupama**, Dittrich Rainer, Technau Gerhard M., Urban Joachim (2001) Successive specification of *Drosophila* neuroblasts NB 6-4 and NB 7-3 depends on interaction of the segment polarity genes wingless, gooseberry and naked cuticle. *Development* **128**:3253-3261 <https://doi.org/10.1242/dev.128.17.3253>
14. **Anderson Nathan**, Lai Sen-Lin, Doe Chris Q. (2025) Vnd and En are expressed in orthogonal stripes and act in a brief competence window to combinatorially specify NB7-1 and its early lineage. *bioRxiv* <https://doi.org/10.1101/2025.09.04.674256>
15. **Isshiki Takako**, Takeichi Masatoshi, Nose Akinao (1997) The role of the msh homeobox gene during *Drosophila* neurogenesis: Implication for the dorsoventral specification of the neuroectoderm. *Development* **124**:3099-3109 <https://doi.org/10.1242/dev.124.16.3099>

16. Weiss Joseph B., Von Ohlen Tonia, Mellerick Dervla M., Dressler Gregory, Doe Chris Q., Scott Matthew P. (1998) Dorsoventral patterning in the Drosophila central nervous system: The intermediate neuroblasts defective homeobox gene specifies intermediate column identity. *Genes & Development* **12**:3591-3602 <https://doi.org/10.1101/gad.12.22.3591>
17. Chu Hsin, Parras Carlos, White Kalpana, Jiménez Fernando (1998) Formation and specification of ventral neuroblasts is controlled by vnd in Drosophila neurogenesis. *Genes & Development* **12**:3613-3624 <https://doi.org/10.1101/gad.12.22.3613>
18. McDonald Jocelyn A., Doe Chris Q. (1997) Establishing neuroblast-specific gene expression in the Drosophila CNS: Hucklebein is activated by Wingless and Hedgehog and repressed by Engrailed and Gooseberry. *Development* **124**:1079-1087 <https://doi.org/10.1242/dev.124.5.1079>
19. Zhang Yu, Li Xin (2024) Development of the Drosophila Optic Lobe. *Cold Spring Harbor Protocols* **2024** <https://doi.org/10.1101/pdb.top108156>
20. Erclik Ted, Li Xin, Courgeon Maximilien, Bertet Claire, Chen Zhenqing, Baumert Ryan, Ng June, Koo Clara, Arain Urfa, Behnia Rudy, *et al.* (2017) Integration of temporal and spatial patterning generates neural diversity. *Nature* **541**:365-370 <https://doi.org/10.1038/nature20794>
21. Gold Katrina S., Brand Andrea H. (2014) Optix defines a neuroepithelial compartment in the optic lobe of the Drosophila brain. *Neural Development* **9**:18 <https://doi.org/10.1186/1749-8104-9-18>
22. Valentino Priscilla, Erclik Ted (2022) Spalt and disco define the dorsal-ventral neuroepithelial compartments of the developing Drosophila medulla. *Genetics* **222**:iyac145 <https://doi.org/10.1093/genetics/iyac145>
23. Apitz Holger, Salecker Iris (2018) Spatio-temporal relays control layer identity of direction-selective neuron subtypes in Drosophila. *Nature Communications* **9**:2295 <https://doi.org/10.1038/s41467-018-04592-z>
24. Doe Chris Q. (2017) Temporal Patterning in the Drosophila CNS. *Annual Review of Cell and Developmental Biology* **33**:219-240 <https://doi.org/10.1146/annurev-cellbio-111315-125210>
25. Maurange Cédric (2020) Temporal patterning in neural progenitors: From Drosophila development to childhood cancers. *Disease Models & Mechanisms* **13**:dmm044883 <https://doi.org/10.1242/dmm.044883>
26. Isshiki Takako, Pearson Bret, Holbrook Scott, Doe Chris Q. (2001) Drosophila Neuroblasts Sequentially Express Transcription Factors which Specify the Temporal Identity of Their Neuronal Progeny. *Cell* **106**:511-521 [https://doi.org/10.1016/S0092-8674\(01\)00465-2](https://doi.org/10.1016/S0092-8674(01)00465-2)
27. Novotny Tanja, Eiselt Regina, Urban Joachim (2002) Hunchback is required for the specification of the early sublineage of neuroblast 7-3 in the Drosophila central nervous system. *Development* **129**:1027-1036 <https://doi.org/10.1242/dev.129.4.1027>
28. Brody Thomas, Odenwald Ward F (2000) Programmed Transformations in Neuroblast Gene Expression during Drosophila CNS Lineage Development. *Developmental Biology* **226**:34-44 <https://doi.org/10.1006/dbio.2000.9829>
29. Konstantinides Nikolaos, Kapuralin Katarina, Fadil Chaimaa, Barboza Luendreo, Satija Rahul, Desplan Claude (2018) Phenotypic Convergence: Distinct Transcription Factors Regulate Common Terminal Features. *Cell* **174**:622-635.e13, <https://doi.org/10.1016/j.cell.2018.05.021>
30. Suzuki Takumi, Kaido Masako, Takayama Rie, Sato Makoto (2013) A temporal mechanism that produces neuronal diversity in the Drosophila visual center. *Developmental Biology* **380**:12-24 <https://doi.org/10.1016/j.ydbio.2013.05.002>
31. Li Xin, Erclik Ted, Bertet Claire, Chen Zhenqing, Voutev Roumen, Venkatesh Srinidhi, Morante Javier, Celik Arzu, Desplan Claude (2013) Temporal patterning of Drosophila medulla neuroblasts controls neural fates. *Nature* **498**:456-462 <https://doi.org/10.1038/nature12319>
32. Konstantinides Nikolaos, Holguera Isabel, Rossi Anthony M., Escobar Aristides, Dudragne Liébaut, Chen Yen-Chung, Tran Thinh N., Jaimes Martínez, Neset Mehmet, Simon Félix, *et al.* (2022) A complete temporal transcription factor series in the fly visual system. *Nature* **604**:316-322

<https://doi.org/10.1038/s41586-022-04564-w>

33. **Zhu Hailun**, Zhao Sihai Dave, Ray Alokandanda, Zhang Yu, Li Xin (2022) A comprehensive temporal patterning gene network in *Drosophila* medulla neuroblasts revealed by single-cell RNA sequencing. *Nature Communications* **13**:1247 <https://doi.org/10.1038/s41467-022-28915-3>
34. **Sato Makoto**, Suzuki Takumi (2022) Cutting edge technologies expose the temporal regulation of neurogenesis in the *Drosophila* nervous system. *Fly* **16**:222-232 <https://doi.org/10.1080/19336934.2022.2073158>
35. **Hasegawa Eri**, Kaido Masako, Takayama Rie, Sato Makoto (2013) Brain-specific-homeobox is required for the specification of neuronal types in the *Drosophila* optic lobe. *Developmental Biology* **377**:90-99 <https://doi.org/10.1016/j.ydbio.2013.02.012>
36. **Hasegawa Eri**, Kitada Yusuke, Kaido Masako, Takayama Rie, Awasaki Takeshi, Tabata Tetsuya, Sato Makoto (2011) Concentric zones, cell migration and neuronal circuits in the *Drosophila* visual center. *Development* **138**:983-993 <https://doi.org/10.1242/dev.058370>
37. **Kanai Makoto I.**, Okabe Masataka, Hiromi Yasushi (2005) Seven-up Controls Switching of Transcription Factors that Specify Temporal Identities of *Drosophila* Neuroblasts. *Developmental Cell* **8**:203-213 <https://doi.org/10.1016/j.devcel.2004.12.014>
38. **Moris-Sanz Marta**, Estacio-Gómez Alicia, Álvarez-Rivero Javier, Díaz-Benjumea Fernando J. (2014) Specification of neuronal subtypes by different levels of Hunchback. *Development* **141**:4366-4374 <https://doi.org/10.1242/dev.113381>
39. **Broadus Julie**, Skeath James B., Spana Eric P., Bossing Torsten, Technau Gerhard, Doe Chris Q. (1995) New neuroblast markers and the origin of the aCC/pCC neurons in the *Drosophila* central nervous system. *Mechanisms of Development* **53**:393-402 [https://doi.org/10.1016/0925-4773\(95\)00454-8](https://doi.org/10.1016/0925-4773(95)00454-8)
40. **Tran Khoa D.**, Doe Chris Q. (2008) Pdm and Castor close successive temporal identity windows in the NB3-1 lineage. *Development (Cambridge, England)* **135**:3491-3499 <https://doi.org/10.1242/dev.024349>
41. **Sen Sonia Q.**, Chanchani Sachin, Southall Tony D, Doe Chris Q (2019) Neuroblast-specific open chromatin allows the temporal transcription factor, Hunchback, to bind neuroblast-specific loci. *eLife* **8**:e44036 <https://doi.org/10.7554/eLife.44036>
42. **Southall Tony D.**, Gold Katrina S., Egger Boris, Davidson Catherine M., Caygill Elizabeth E., Marshall Owen J., Brand Andrea H. (2013) Cell-Type-Specific Profiling of Gene Expression and Chromatin Binding without Cell Isolation: Assaying RNA Pol II Occupancy in Neural Stem Cells. *Developmental Cell* **26**:101-112 <https://doi.org/10.1016/j.devcel.2013.05.020>
43. **The modENCODE Consortium**, Roy Sushmita, Ernst Jason, Kharchenko Peter V., Kheradpour Pouya, Negre Nicolas, Eaton Matthew L., Landolin Jane M., Bristow Christopher A., Ma Lijia, *et al.* (2010) Identification of Functional Elements and Regulatory Circuits by *Drosophila* modENCODE. *Science* **330**:1787-1797 <https://doi.org/10.1126/science.1198374>
44. **Baumgartner S.**, Bopp D., Burri M., Noll M. (1987) Structure of two genes at the gooseberry locus related to the paired gene and their spatial expression during *Drosophila* embryogenesis. *Genes & Development* **1**:1247-1267 <https://doi.org/10.1101/gad.1.10.1247>
45. **Bonneaud Nathalie**, Layalle Sophie, Colomb Sophie, Jourdan Christophe, Ghysen Alain, Severac Dany, Dantec Christelle, Nègre Nicolas, Maschat Florence (2017) Control of nerve cord formation by Engrailed and Gooseberry-Neuro: A multi-step, coordinated process. *Developmental Biology* **432**:273-285 <https://doi.org/10.1016/j.ydbio.2017.10.018>
46. **Li X.**, Noll M. (1993) Role of the gooseberry gene in *Drosophila* embryos: Maintenance of wingless expression by a wingless–gooseberry autoregulatory loop. *The EMBO Journal* **12**:4499-4509 <https://doi.org/10.1002/j.1460-2075.1993.tb06139.x>
47. **Gutjahr Thomas**, Patel Nipam H., Li Xuelin, Goodman Corey S., Noll Markus (1993) Analysis of the gooseberry locus in *Drosophila* embryos: Gooseberry determines the cuticular pattern and activates gooseberry neuro. *Development* **118**:21-31 <https://doi.org/10.1242/dev.118.1.21>

48. **BioGRID consortium** (2025) Biogrid: Result summary for gsb (dmel_cg3388). Accessed YYYY-MM-DD <https://thebiogrid.org/63573>
49. **BioGRID Consortium** (2025) Biogrid: Result summary for en (dmel_cg9015). Accessed YYYY-MM-DD <https://thebiogrid.org/62028>
50. **Serrano Nuria**, Brock Hugh W., Maschat Florence (1997) β 3-tubulin is directly repressed by the Engrailed protein in Drosophila. *Development* **124**:2527-2536 <https://doi.org/10.1242/dev.124.13.2527>
51. **Renault Andrew D.** (2012) Vasa is expressed in somatic cells of the embryonic gonad in a sexspecific manner in Drosophila melanogaster. *Biology Open* **1**:1043-1048 <https://doi.org/10.1242/bio.20121909>
52. **Baumgardt Magnus**, Karlsson Daniel, Terriente Javier, Díaz-Benjumea Fernando J., Thor Stefan (2009) Neuronal Subtype Specification within a Lineage by Opposing Temporal Feed-Forward Loops. *Cell* **139**:969-982 <https://doi.org/10.1016/j.cell.2009.10.032>
53. **Gabilondo Hugo**, Stratmann Johannes, Rubio-Ferrera Irene, Millán-Crespo Irene, Contero-García Patricia, Bahrampour Shahrzad, Thor Stefan, Benito-Sipos Jonathan (2016) Neuronal Cell Fate Specification by the Convergence of Different Spatiotemporal Cues on a Common Terminal Selector Cascade. *PLOS Biology* **14**:e1002450 <https://doi.org/10.1371/journal.pbio.1002450>
54. **Nevil Markus**, Gibson Tyler J., Bartolutti Constantine, Iyengar Anusha, Harrison Melissa M. (2020) Establishment of chromatin accessibility by the conserved transcription factor Grainy head is developmentally regulated. *Development* **147**:dev185009 <https://doi.org/10.1242/dev.185009>
55. **Barral Amandine**, Zaret Kenneth S. (2024) Pioneer factors: Roles and their regulation in development. *Trends in Genetics* **40**:134-148 <https://doi.org/10.1016/j.tig.2023.10.007>
56. **Davis Gregory K.**, D'Alessio Joseph A., Patel Nipam H. (2005) Pax3/7 genes reveal conservation and divergence in the arthropod segmentation hierarchy. *Developmental Biology* **285**:169-184 <https://doi.org/10.1016/j.ydbio.2005.06.014>
57. **Xue L**, Noll M (1996) The functional conservation of proteins in evolutionary alleles and the dominant role of enhancers in evolution. *The EMBO Journal* **15**:3722-3731
58. **Mayran Alexandre**, Khetchoumian Konstantin, Hariri Fadi, Pastinen Tomi, Gauthier Yves, Balsalobre Aurelio, Drouin Jacques (2018) Pioneer factor Pax7 deploys a stable enhancer repertoire for specification of cell fate. *Nature Genetics* **50**:259-269 <https://doi.org/10.1038/s41588-017-0035-2>
59. **Mayran Alexandre**, Sochodolsky Kevin, Khetchoumian Konstantin, Harris Juliette, Gauthier Yves, Bemmo Amandine, Balsalobre Aurelio, Drouin Jacques (2019) Pioneer and nonpioneer factor cooperation drives lineage specific chromatin opening. *Nature Communications* **10**:3807 <https://doi.org/10.1038/s41467-019-11791-9>
60. **Budry Lionel**, Balsalobre Aurélio, Gauthier Yves, Khetchoumian Konstantin, L'Honoré Aurore, Vallette Sophie, Brue Thierry, Figarella-Branger Dominique, Meij Björn, Drouin Jacques (2012) The selector gene Pax7 dictates alternate pituitary cell fates through its pioneer action on chromatin remodeling. *Genes & Development* **26**:2299-2310 <https://doi.org/10.1101/gad.200436.112>
61. **McKinnell Iain W.**, Ishibashi Jeff, Le Grand Fabien, Punch Vincent G. J., Addicks Gregory C., Greenblatt Jack F., Dilworth F. Jeffrey, Rudnicki Michael A. (2008) Pax7 activates myogenic genes by recruitment of a histone methyltransferase complex. *Nature Cell Biology* **10**:77-84 <https://doi.org/10.1038/ncb1671>
62. **Diao Yarui**, Guo Xing, Li Yanfeng, Sun Kun, Lu Leina, Jiang Lei, Fu Xinrong, Zhu Han, Sun Hao, Wang Huating, *et al.* (2012) Pax3/7BP Is a Pax7- and Pax3-Binding Protein that Regulates the Proliferation of Muscle Precursor Cells by an Epigenetic Mechanism. *Cell Stem Cell* **11**:231-241 <https://doi.org/10.1016/j.stem.2012.05.022>
63. **Kawabe Yoh-ichi**, Wang Yu Xin, McKinnell Iain W., Bedford Mark T., Rudnicki Michael A. (2012) Carm1 Regulates Pax7 Transcriptional Activity through MLL1/2 Recruitment during Asymmetric Satellite Stem Cell Divisions. *Cell Stem Cell* **11**:333-345 <https://doi.org/10.1016/j.stem.2012.07.001>

64. Gryder Berkley E., Yohe Marielle E., Chou Hsien-Chao, Zhang Xiaohu, Marques Joana, Wachtel Marco, Schaefer Beat, Sen Nirmalya, Song Young, Gualtieri Alberto, *et al.* (2017) PAX3-FOXO1 Establishes Myogenic Super Enhancers and Confers BET Bromodomain Vulnerability. *Cancer Discovery* **7**:884-899 <https://doi.org/10.1158/2159-8290.CD-16-1297>
65. Tolkunova Elena N., Fujioka Miki, Kobayashi Masatomo, Deka Deepali, Jaynes James B. (1998) Two Distinct Types of Repression Domain in Engrailed: One Interacts with the Groucho Corepressor and Is Preferentially Active on Integrated Target Genes. *Molecular and Cellular Biology* **18**:2804-2814 <https://doi.org/10.1128/mcb.18.5.2804>
66. Copley Richard R. (2005) The EH1 motif in metazoan transcription factors. *BMC Genomics* **6**:169 <https://doi.org/10.1186/1471-2164-6-169>
67. Chen G., Fernandez J., Mische S., Courey A. J. (1999) A functional interaction between the histone deacetylase Rpd3 and the corepressor Groucho in Drosophila development. *Genes & Development* **13**:2218-2230 <https://doi.org/10.1101/gad.13.17.2218>
68. Kaul Aamna, Schuster Eugene, Jennings Barbara H. (2014) The Groucho Co-repressor Is Primarily Recruited to Local Target Sites in Active Chromatin to Attenuate Transcription. *PLOS Genetics* **10**:e1004595 <https://doi.org/10.1371/journal.pgen.1004595>
69. Luo Lichao, Siah Chia Keng, Cai Yu (2017) Engrailed acts with Nejire to control decapentaplegic expression in the Drosophila ovarian stem cell niche. *Development* **144**:3224-3231 <https://doi.org/10.1242/dev.145474>
70. Alexandre Cyrille, Vincent Jean-Paul (2003) Requirements for transcriptional repression and activation by Engrailed in Drosophila embryos. *Development* **130**:729-739 <https://doi.org/10.1242/dev.00286>
71. Maschat Florence, Serrano Nuria, Randsholt Neel B., Géraud Gérard (1998) Engrailed and polyhomeotic interactions are required to maintain the A/P boundary of the Drosophila developing wing. *Development* **125**:2771-2780 <https://doi.org/10.1242/dev.125.15.2771>
72. Urbach Rolf, Technau Gerhard M. (2004) Neuroblast formation and patterning during early brain development in Drosophila. *BioEssays* **26**:739-751 <https://doi.org/10.1002/bies.20062>
73. Sen Sonia, Biagini Silvia, Reichert Heinrich, VijayRaghavan K. (2014) Orthodenticle is required for the development of olfactory projection neurons and local interneurons in Drosophila. *Biology Open* **3**:711-717 <https://doi.org/10.1242/bio.20148524>
74. Frith Thomas J.R., Briscoe James, Boezio Giulia L.M. (2024) Chapter Five - From signalling to form: The coordination of neural tube patterning. In: Mallo Moises (Ed). *Vertebrate Pattern Formation, volume 159 of Current Topics in Developmental Biology* Academic Press. pp. 168-231 <https://doi.org/10.1016/bs.ctdb.2023.11.004>
75. Sagner Andreas, Zhang Isabel, Watson Thomas, Lazaro Jorge, Melchionda Manuela, Briscoe James (2021) A shared transcriptional code orchestrates temporal patterning of the central nervous system. *PLOS Biology* **19**:e3001450 <https://doi.org/10.1371/journal.pbio.3001450>
76. Zhang Isabel, Boezio Giulia L. M., Cornwall-Scoones Jake, Frith Thomas, Finnie Elizabeth, Luo Junyi, Jiang Ming, Howell Michael, Lovell-Badge Robin, Sagner Andreas, *et al.* (2025) The cis-regulatory logic integrating spatial and temporal patterning in the vertebrate neural tube. *Developmental Cell* **0** <https://doi.org/10.1016/j.devcel.2025.06.029>
77. Port Phillip, Chen Hui-Min, Lee Tzumin, Bullock Simon L. (2014) Optimized CRISPR/Cas tools for efficient germline and somatic genome engineering in Drosophila. *Proceedings of the National Academy of Sciences* **111**:E2967-E2976 <https://doi.org/10.1073/pnas.1405500111>
78. Marshall Owen J., Southall Tony D., Cheetham Seth W., Brand Andrea H. (2016) Cell-typespecific profiling of protein–DNA interactions without cell isolation using targeted DamID with next-generation sequencing. *Nature Protocols* **11**:1586-1598 <https://doi.org/10.1038/nprot.2016.084>
79. Langmead Ben, Salzberg Steven L. (2012) Fast gapped-read alignment with Bowtie 2. *Nature Methods* **9**:357-359 <https://doi.org/10.1038/nmeth.1923>

80. Li Heng, Handsaker Bob, Wysoker Alec, Fennell Tim, Ruan Jue, Homer Nils, Marth Gabor, Abecasis Goncalo, Durbin Richard (2009) and 1000 Genome Project Data Processing Subgroup. The Sequence Alignment/Map format and SAMtools. *Bioinformatics* **25**:2078-2079 <https://doi.org/10.1093/bioinformatics/btp352>
 81. Breese Marcus R., Liu Yunlong (2013) NGSUtils: A software suite for analyzing and manipulating next-generation sequencing datasets. *Bioinformatics* **29**:494-496 <https://doi.org/10.1093/bioinformatics/bts731>
 82. Ramírez Fidel, Ryan Devon P, Grüning Björn, Bhardwaj Vivek, Kilpert Fabian, Richter Andreas S, Heyne Steffen, Dündar Friederike, Manke Thomas (2016) deepTools2: A next generation web server for deep-sequencing data analysis. *Nucleic Acids Research* **44**:W160-W165 <https://doi.org/10.1093/nar/gkw257>
 83. Zhang Yong, Liu Tao, Meyer Clifford A., Eeckhoutte Jérôme, Johnson David S., Bernstein Bradley E., Nusbaum Chad, Myers Richard M., Brown Myles, Li Wei, *et al.* (2008) Model-based Analysis of ChIP-Seq (MACS). *Genome Biology* **9**:R137 <https://doi.org/10.1186/gb-2008-9-9-r137>
 84. Stark Rory, Brown Gordon (2011) DiffBind: differential binding analysis of ChIP-Seq peak data. *Bioconductor Vignette*.
 85. Ross-Innes Caryn S., Stark Rory, Teschendorff Andrew E., Holmes Kelly A., Ali H. Raza, Dunning Mark J., Brown Gordon D., Gojis Ondrej, Ellis Ian O., Green Andrew R., *et al.* (2012) Differential oestrogen receptor binding is associated with clinical outcome in breast cancer. *Nature* **481**:389-393 <https://doi.org/10.1038/nature10730>
 86. Robinson James T., Thorvaldsdóttir Helga, Winckler Wendy, Guttman Mitchell, Lander Eric S., Getz Gad, Mesirov Jill P. (2011) Integrative genomics viewer. *Nature Biotechnology* **29**:24-26 <https://doi.org/10.1038/nbt.1754>
 87. Thorvaldsdóttir Helga, Robinson James T., Mesirov Jill P. (2013) Integrative Genomics Viewer (IGV): High-performance genomics data visualization and exploration. *Briefings in Bioinformatics* **14**:178-192 <https://doi.org/10.1093/bib/bbs017>
 88. Schindelin Johannes, Arganda-Carreras Ignacio, Frise Erwin, Kaynig Verena, Longair Mark, Pietzsch Tobias, Preibisch Stephan, Rueden Curtis, Saalfeld Stephan, Schmid Benjamin, *et al.* (2012) Fiji: An open-source platform for biological-image analysis. *Nature Methods* **9**:676-682 <https://doi.org/10.1038/nmeth.2019>
- Ayanthi Bhattacharya, Sonia Q Sen (2025) Chromatin priming and Hunchback recruitment integrate spatial and temporal cues in *Drosophila* neuroblasts. NCBI Sequence Read Archive. ID PRJNA1372925 <https://www.ncbi.nlm.nih.gov/sra/PRJNA1372925>

Peer reviews

Reviewer #1 (Public review):

Summary:

It has long been known that *Drosophila* embryonic ventral nerve cord neuroblasts incorporate both spatial and temporal transcription factor expression to generate 30 distinct neuroblasts and lineages per hemisegment. This manuscript aims to elucidate the mechanism by which this integration of spatial and temporal transcription factors occurs through "direct regulation" or "epigenetic regulation". Direct regulation is defined as both spatial and temporal factors binding to open chromatin and working together to dictate specific lineages. Epigenetic regulation is defined as a spatial factor priming the chromatin in a neuroblast-specific manner to allow for the integration of temporal factors to generate specific lineages. The authors conclude that there is a two-step model in which a spatial transcription factor code "primes" the chromatin in terms of accessibility and then recruits temporal factors to ensure lineage-specific enhancer activation.

Strengths:

The authors tested two models, "direct regulation" vs "epigenetic regulation" in a well-defined pool of neural stem cells during normal development.

Weaknesses:

The data in this study cannot clearly substantiate these two models.

Overall, there are a number of issues that are inconsistent and not supportive of the model proposed in this manuscript. Firstly, there is no evidence of pioneer factor activity in any of the NB lineages described - i.e., any changes in chromatin accessibility being shown over time. The authors must show chromatin conformation changes during the window of spatial transcription factor expression in order to convince the readers of this phenomenon. Secondly, the phenotypic data do not align with the sequencing data - the story would be more cohesive if the sequencing data and phenotypic data were in the same NB subtypes. On one hand, we are shown that Gsb misexpression induces loss of chromatin accessibility in NB 7-4, however in the widespread loss model, we are not shown a phenotype in these NB7-4 - which suggest that the chromatin accessibility at these sites (sites that have already been distinguished as SoIs for that NB subtype) does not play an important role in distinguishing NB 7-4 identity. However, the authors report loss of NB3-5 identity but have no evidence as to how the chromatin has changed (or if it has at all) in that subtype, leaving the readers to wonder how the loss of identity occurred.

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Reviewer #2 (Public review):

Summary:

This article by Bhattacharya et al. investigates how neural stem cells (NSCs, NBs) in *Drosophila* integrate spatial and temporal cues to activate neuron-specific terminal selector (TS) genes. Prior to this work, it was understood that NSCs utilize spatial transcription factors (STFs) and temporal transcription factors (TTFs) to determine lineage identity and birth order, but the mechanisms of integration were not fully elucidated. The authors employed chromatin profiling techniques to analyze the binding of STFs and TTFs in two specific neuroblast lineages, NB5-6 and NB7-4. They found that Gsb (an STF) binds both accessible and less-accessible chromatin in NB5-6, while En (another STF) binds only to pre-accessible chromatin in NB7-4. The findings support an "STF code" where the combination of pioneer and non-pioneer spatial factors, along with temporal factors, triggers neuroblast-specific enhancer activation and determines lineage identity.

Strengths:

The experiments are well-executed, the interpretations are generally sound, and the figures are clear and elegant. However, some conclusions are drawn too broadly without essential functional data. Therefore, additional work is needed to more effectively convey the central message.

Weaknesses:

(1) Integration of TaDa and functional data on Gsb for the STF model

The authors demonstrate that TaDa profiling maps Gsb binding across the genome and identifies candidate chromatin-priming sites in NB5-6. Gsb LOF/GOF experiments reveal effects on NB identity. Combining TaDa data with LOF and GOF analyses indicates that Gsb influences NB5-6 specification by binding to both open and relatively closed chromatin, helping maintain NB5-6 identity while limiting NB3-5 fate.

However, the study does not establish a direct link between specific LOF/GOF phenotypes and particular genomic targets. For instance, analyzing Gsb occupancy at lineage-specific identity factors or terminal selector genes (such as *Lbe*, *Ap*, or *Eya* for NB5-6; and *Ems*, etc., for NB3-5) in wild-type and manipulated conditions (Gsb misexpression) would directly connect chromatin binding to the regulation of fate determinants. These investigations would strengthen the mechanistic connection between the correlative TaDa profiles and the observed identity changes, supporting the idea that Gsb functions as a context-dependent chromatin-priming factor within the STF code, rather than as a generic transcription factor.

(2) Gsb misexpression reveals bidirectional chromatin remodelling

Experiments with ectopic Gsb expression demonstrate bidirectional chromatin remodeling in NB7-4, showing decreases in accessibility at some binding sites and increases at others. While the authors show that Gsb can disrupt chromatin upon misexpression, interpreting its "pioneer-like" or chromatin-priming activity is complex due to several factors: the misexpression occurs in a non-native lineage, the direct versus indirect effects rely on whole-embryo Dam-Gsb peaks instead of NB7-4-specific binding, and heat-shock-induced chromatin changes are not fully accounted for. These issues make it challenging to definitively determine Gsb's role in chromatin priming.

A complementary approach would be to perform Gsb knockdown/loss-of-function in its native NB5-6 lineage and profile chromatin accessibility (TaDa or CATaDa). This would allow a cleaner, more physiologically relevant assessment of Gsb's contribution to priming, SoI establishment, and Hb recruitment. Such an experiment would strengthen the causal link between Gsb occupancy and chromatin state and clarify whether Gsb truly acts as a context-dependent pioneer in vivo, rather than producing indirect effects due to ectopic misexpression.

(3) En is not a pioneer factor

The authors conclude that Engrailed (*En*) is not a pioneer factor, based on the observation that *En* binding correlates with accessible chromatin and that *En* is not enriched at NB5-6-specific SOIs. However, this conclusion is not sufficiently supported by the functional data.

First, the absence of *En* binding at NB5-6-specific SOIs does not necessarily indicate an inability to engage closed chromatin. These regions were not selected for the presence of *En* consensus motifs, so their lack of occupancy may simply reflect the absence of *En* binding motifs rather than a lack of pioneering capacity. A systematic motif analysis at NB5-6-specific SOIs is needed to determine whether *En* binding sites are present but unoccupied.

Second, the claim that *En* lacks pioneer activity relies solely on a single steady-state TaDa/DamID occupancy assay at one developmental stage. Because pioneer factor interactions can be transient, low-affinity, and stage-specific, such binding may not be detected by TaDa, which also depends on local GATC density and methylation kinetics and may yield false negatives. Given these technical limitations, the absence of *En* binding at less accessible regions does not definitively rule out a priming role.

In the absence of direct functional assays (*En* LOF/GOF), the authors should explicitly acknowledge these technical and conceptual limitations and tone down the claim that "*En* lacks pioneer activity".

(4) Clarity of STF-code Model and Central Message

The manuscript begins by presenting two models, direct and epigenetic, but the central takeaway of the paper is not clear. Specifically, the nuanced roles of the spatial factors Gsb and *En* as chromatin-priming versus stabilizing/effector factors within an STF code, and the resulting division of labor, are not clearly illustrated. The distinction between Gsb as a

chromatin-priming factor and En as a cofactor-dependent activator/stabilizer should be explicitly presented in a stepwise model for better clarity. The authors could strengthen this by providing a schematic with two sequential stages illustrating how neuroblast identity factors (STF code) change chromatin states to drive lineage-specific enhancer activation. The schematic can be shown from the neuroectoderm to individual NB lineages to make it more panoramic.

(5) Identification of Priming Factors in NB7-4

While the authors suggest that an unknown priming factor might be responsible for establishing sites of integration in NB7-4, they do not identify or explore potential candidates for this role. Further investigation into what factors might be involved in chromatin priming in NB7-4 could provide a more complete understanding of the mechanisms at play.

(6) Functional Validation of STF Code Components

The study proposes an STF code for each neuroblast lineage, but the specific components of these codes, beyond Gsb and En, are not fully explored. Identifying and validating additional factors that contribute to the STF code in each lineage could strengthen the conclusions.

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Author Response:

eLife assessment:

The study provides an important advance towards understanding how spatial and temporal transcriptional programs are integrated to regulate lineage-specific chromatin and enhancer activation. The functional evidence is currently incomplete, but the current data provide a solid correlative and conceptual foundation. Functional experiments directly linking Gsb occupancy to chromatin state and regulation of some lineage-specific targets would further strengthen the causal interpretation of the model. Clarifying the scope of conclusions and explicitly acknowledging the technical limitations of current chromatin assays would provide a more balanced interpretation of the manuscript.

We thank the reviewers and editors for their comments on our manuscript. We address here the concerns raised by them.

Public Reviews:

Reviewer #1 (Public review):

Summary:

It has long been known that Drosophila embryonic ventral nerve cord neuroblasts incorporate both spatial and temporal transcription factor expression to generate 30 distinct neuroblasts and lineages per hemisegment. This manuscript aims to elucidate the mechanism by which this integration of spatial and temporal transcription factors occurs through "direct regulation" or "epigenetic regulation". Direct regulation is defined as both spatial and temporal factors binding to open chromatin and working together to dictate specific lineages. Epigenetic regulation is defined as a spatial factor priming the chromatin in a neuroblast-specific manner to allow for the integration of temporal factors to generate specific lineages. The authors conclude that there is a two-step model in which a spatial transcription factor code "primes" the chromatin in terms of accessibility and then recruits temporal factors to ensure lineage-specific enhancer activation.

We thank the reviewer for this clear and succinct summary and for accurately capturing the central idea of the model we propose. In particular, we appreciate that the reviewer highlights the distinction between the previously proposed “direct regulation” and “epigenetic regulation” models, which our work suggests may operate together within neuroblast lineages through a combinatorial spatial transcription factor code.

Strengths:

The authors tested two models, "direct regulation" vs "epigenetic regulation" in a well-defined pool of neural stem cells during normal development.

We thank the reviewer for recognizing this aspect of the study.

Weaknesses:

The data in this study cannot clearly substantiate these two models.

Overall, there are a number of issues that are inconsistent and not supportive of the model proposed in this manuscript. Firstly, there is no evidence of pioneer factor activity in any of the NB lineages described - i.e., any changes in chromatin accessibility being shown over time. The authors must show chromatin conformation changes during the window of spatial transcription factor expression in order to convince the readers of this phenomenon.

Thank you for raising this point. In most studies, pioneer or chromatin-priming activity is inferred from a transcription factor’s ability to bind regions of relatively low accessibility and to remodel chromatin upon perturbation, rather than from direct developmental time-course measurements of chromatin accessibility.

In our study we provide two lines of evidence consistent with such activity. First, TaDa profiling shows that Gsb occupies both accessible loci and regions that are relatively less accessible in NB5-6. Second, ectopic expression of Gsb in the non-cognate NB7-4 lineage results in clear chromatin remodelling, with loci both gaining and losing accessibility (Fig. 6). These perturbation experiments demonstrate that Gsb is sufficient to alter chromatin accessibility *in vivo* and therefore support a chromatin-priming role for it.

We agree that a developmental time-course would be very informative. The difficulty is that, in this system, the relevant sequence unfolds extremely rapidly and across two different cellular contexts. Spatial transcription factors such as Gsb are expressed in the neuroectoderm, neuroblasts are then specified and delaminate, and Hb expression begins almost immediately after NB formation — on the order of minutes to tens of minutes. Before delamination there is no neuroblast to target with NB-specific drivers, and once the NB forms the temporal program is already underway. More generally, resolving chromatin accessibility changes across this transition would require temporally precise profiling at very high resolution *in vivo*, likely with live or near-live methods, and is not feasible with the Dam-based lineage-restricted approaches currently available.

Secondly, the phenotypic data do not align with the sequencing data - the story would be more cohesive if the sequencing data and phenotypic data were in the same NB subtypes. On one hand, we are shown that Gsb misexpression induces loss of chromatin accessibility in NB 7-4, however in the widespread loss model, we are not shown a phenotype in these NB7-4 - which suggest that the chromatin accessibility at these sites (sites that have already been distinguished as SoIs for that NB subtype) does not play an important role in distinguishing NB 7-4 identity. However, the authors report loss of NB3-5 identity but have no evidence as to how the chromatin has changed (or if it has at all) in that subtype, leaving the readers to wonder how the loss of identity occurred

Thank you for raising this point regarding the alignment between the chromatin and phenotypic analyses. The reviewer's comment made us realise that the rationale for these experiments may not have been sufficiently clear in the original manuscript and could therefore be perceived as misaligned. We therefore explain the logic of the experimental design here and will edit the manuscript in the revision to clarify this point for readers.

The chromatin experiments were designed to test whether Gsb is capable of remodelling chromatin when introduced into a non-cognate lineage. For this purpose, NB7-4 provided a suitable lineage with clean genetic access for TaDa/CATaDa experiments, allowing us to assess whether ectopic Gsb expression can alter chromatin accessibility in vivo.

The functional role of Gsb, however, was examined within the spatial domain in which it is normally expressed. We knocked-down Gsb broadly and early in development and assayed its effects on NB5-6. Consistent with its established role in row-5/6 patterning, reduction of Gsb disrupted the specification of NB5-6 identity. In the converse experiment, broad misexpression of Gsb led to a partial expansion of NB5-6 markers. Because spatial patterning in the ventral nerve cord is organized into mutually exclusive row identities, changes in NB5-6 specification can be accompanied by reciprocal effects in neighbouring lineages. In our experiments, this is reflected in changes in markers of adjacent identities, particularly NB3-5. For this reason, NB3-5 markers provide a sensitive and informative readout of altered NB5-6 specification in the phenotypic analyses.

We recognize that this point may not have been clear in the original manuscript. To avoid similar confusion for readers, we will make this reasoning explicitly clear in the revision.

Reviewer #2 (Public review):

Summary:

This article by Bhattacharya et al. investigates how neural stem cells (NSCs, NBs) in Drosophila integrate spatial and temporal cues to activate neuron-specific terminal selector (TS) genes. Prior to this work, it was understood that NSCs utilize spatial transcription factors (STFs) and temporal transcription factors (TTFs) to determine lineage identity and birth order, but the mechanisms of integration were not fully elucidated. The authors employed chromatin profiling techniques to analyze the binding of STFs and TTFs in two specific neuroblast lineages, NB5-6 and NB7-4. They found that Gsb (an STF) binds both accessible and less-accessible chromatin in NB5-6, while En (another STF) binds only to pre-accessible chromatin in NB7-4. The findings support an "STF code" where the combination of pioneer and non-pioneer spatial factors, along with temporal factors, triggers neuroblast-specific enhancer activation and determines lineage identity.

We appreciate the reviewer's careful summary of our findings and their clear articulation of the STF-code framework that emerges from the work.

Strengths:

The experiments are well-executed, the interpretations are generally sound, and the figures are clear and elegant. However, some conclusions are drawn too broadly without essential functional data. Therefore, additional work is needed to more effectively convey the central message.

We thank the reviewer for their positive assessment of the experiments, interpretation, and figures, and we respond to their specific concerns below.

Weaknesses:

(1) Integration of TaDa and functional data on Gsb for the STF model

The authors demonstrate that TaDa profiling maps Gsb binding across the genome and identifies candidate chromatin-priming sites in NB5-6. Gsb LOF/GOF experiments reveal effects on NB identity. Combining TaDa data with LOF and GOF analyses indicates that Gsb influences NB5-6 specification by binding to both open and relatively closed chromatin, helping maintain NB5-6 identity while limiting NB3-5 fate.

However, the study does not establish a direct link between specific LOF/GOF phenotypes and particular genomic targets. For instance, analyzing Gsb occupancy at lineage-specific identity factors or terminal selector genes (such as Lbe, Ap, or Eya for NB5-6; and Ems, etc., for NB3-5) in wild-type and manipulated conditions (Gsb misexpression) would directly connect chromatin binding to the regulation of fate determinants. These investigations would strengthen the mechanistic connection between the correlative TaDa profiles and the observed identity changes, supporting the idea that Gsb functions as a context-dependent chromatin-priming factor within the STF code, rather than as a generic transcription factor.

We thank the reviewer for this very helpful suggestion. We agree that illustrating how the TaDa binding profiles relate to known lineage determinants will help connect the genome-wide chromatin data to the developmental phenotypes. In the revision therefore, we will examine Gsb occupancy at several genes associated with NB5-6 and NB3-5 identity (including Lbe, Ap, Eya, and Ems).

(2) Gsb misexpression reveals bidirectional chromatin remodelling

Experiments with ectopic Gsb expression demonstrate bidirectional chromatin remodeling in NB7-4, showing decreases in accessibility at some binding sites and increases at others. While the authors show that Gsb can disrupt chromatin upon misexpression, interpreting its "pioneer-like" or chromatin-priming activity is complex due to several factors: the misexpression occurs in a non-native lineage, the direct versus indirect effects rely on whole-embryo Dam-Gsb peaks instead of NB7-4-specific binding, and heat-shock-induced chromatin changes are not fully accounted for. These issues make it challenging to definitively determine Gsb's role in chromatin priming.

A complementary approach would be to perform Gsb knockdown/loss-of-function in its native NB5-6 lineage and profile chromatin accessibility (TaDa or CATaDa). This would allow a cleaner, more physiologically relevant assessment of Gsb's contribution to priming, SoI establishment, and Hb recruitment. Such an experiment would strengthen the causal link between Gsb occupancy and chromatin state and clarify whether Gsb truly acts as a context-dependent pioneer in vivo, rather than producing indirect effects due to ectopic misexpression.

We thank the reviewer for this thoughtful comment. We agree that the ectopic Gsb misexpression experiment in NB7-4 should be interpreted as a test of chromatin-remodelling capacity rather than as a fully physiological assay of Gsb function in its native NB5-6 context. At the same time, we note that ectopic expression in a non-native lineage is a standard approach used to assess pioneering or chromatin-remodelling capacity, precisely because it tests whether a factor can alter chromatin outside its endogenous setting. In the revision, we will explicitly discuss this distinction.

We also agree that NB7-4-specific Gsb occupancy under misexpression would provide a cleaner distinction between direct and indirect effects. In the current manuscript, we infer likely direct effects from overlap with whole-embryo Gsb Dam profiles: loci that lose accessibility upon Gsb misexpression overlap whole-embryo Gsb binding, whereas loci that gain accessibility generally do not. We interpret this as support for the idea that decreased

accessibility is more likely to reflect direct Gsb action, whereas increased accessibility is more likely to be indirect. We will clarify this logic in the revision.

Regarding the reviewer's suggestion of profiling chromatin accessibility after Gsb loss in native NB5-6, we completely agree that this would be an important complementary experiment. However, this experiment is not currently possible in our system. Gsb is required before NB specification/delamination, whereas available NB5-6 Gal4 drivers turn on only after this stage, precluding the use of RNAi. Early mutant analysis is also technically difficult because homozygous mutant embryos cannot be readily identified at the required stage, and the TaDa/CATaDa approach in this system requires large amounts of input material collected during the very short Hb window. We also tested an early CRISPR-based strategy using maternally contributed Cas9, but in this context the NB5-6 driver is lost, preventing TaDa/CATaDa profiling. We will therefore revise the manuscript to acknowledge that the current misexpression data support chromatin-remodelling capacity and are consistent with context-dependent priming, while not definitively establishing endogenous priming activity in NB5-6.

(3) En is not a pioneer factor

The authors conclude that Engrailed (En) is not a pioneer factor, based on the observation that En binding correlates with accessible chromatin and that En is not enriched at NB5-6-specific SOIs. However, this conclusion is not sufficiently supported by the functional data.

We thank the reviewer for raising this point. We agree that, in several places, our wording was stronger than warranted by the data. For example, we stated that this pattern “argues against a pioneer role for En” and that the results “indicate that En does not act as a pioneer factor.” We agree that these statements are too definitive given the current evidence. Below, we address each of the reviewer's specific concerns and explain the reasoning behind our original interpretation.

First, the absence of En binding at NB5-6-specific SOIs does not necessarily indicate an inability to engage closed chromatin. These regions were not selected for the presence of En consensus motifs, so their lack of occupancy may simply reflect the absence of En binding motifs rather than a lack of pioneering capacity. A systematic motif analysis at NB5-6-specific SOIs is needed to determine whether En binding sites are present but unoccupied.

We agree that the absence of En binding at NB5-6-specific SOIs alone would not be sufficient to infer a lack of pioneering activity, particularly if these loci do not contain En consensus motifs. That observation was only the starting point for our interpretation. Our reasoning was based on several additional lines of evidence from the genome-wide analysis:

- (1) When we examined En binding genome-wide, we consistently found that En occupancy in NB7-4 is restricted to regions of accessible chromatin.
- (2) Loci that are less accessible in NB7-4 show no detectable En occupancy.
- (3) Accessibility is strongly predictive of En binding: chromatin accessibility is markedly higher at En-bound loci than at En-unbound loci.

Taken together, these patterns suggested to us that En binding in this lineage occurs primarily at pre-accessible chromatin rather than at less accessible regions that would require priming.

Our interpretation was also guided by the broader literature. To our knowledge, neither *Drosophila* Engrailed nor its vertebrate homologues (EN1/EN2) have been reported to bind

nucleosome-occluded DNA or initiate chromatin opening, which further informed our original interpretation.

That said, we agree with the reviewer that these observations are suggestive rather than definitive. We will therefore temper the language throughout the manuscript so that we do not make categorical claims about En lacking pioneer activity. We will also perform the suggested motif analysis at NB5-6-specific SOIs to determine whether En binding motifs are present at these loci, which should help clarify whether the lack of En occupancy reflects motif availability or chromatin state.

Second, the claim that En lacks pioneer activity relies solely on a single steady-state TaDa/DamID occupancy assay at one developmental stage. Because pioneer factor interactions can be transient, low-affinity, and stage-specific, such binding may not be detected by TaDa, which also depends on local GATC density and methylation kinetics and may yield false negatives. Given these technical limitations, the absence of En binding at less accessible regions does not definitively rule out a priming role.

We take the reviewer's point that our data cannot definitively rule out En as a pioneer. At the same time, it may be useful to clarify that TaDa is not a snapshot assay. Because Dam-mediated methylation accumulates over time while the fusion protein is expressed, even weak or transient interactions can leave a detectable signal when averaged across many cells and across the duration of the expression window.

This cumulative nature of the assay is why our consistent observation of strong enrichment of En at accessible loci, and no detectable enrichment at less accessible regions across the genome, led us to infer that En binding in NB7-4 is strongly conditioned on chromatin accessibility. We nevertheless agree that this does not definitively exclude rare or transient interactions below the detection threshold of the assay, and we will temper the language in the manuscript accordingly.

In the absence of direct functional assays (En LOF/GOF), the authors should explicitly acknowledge these technical and conceptual limitations and tone down the claim that "En lacks pioneer activity".

Yes, we will do that!

(4) Clarity of STF-code Model and Central Message

The manuscript begins by presenting two models, direct and epigenetic, but the central takeaway of the paper is not clear. Specifically, the nuanced roles of the spatial factors Gsb and En as chromatin-priming versus stabilizing/effector factors within an STF code, and the resulting division of labor, are not clearly illustrated. The distinction between Gsb as a chromatin-priming factor and En as a cofactor-dependent activator/stabilizer should be explicitly presented in a stepwise model for better clarity. The authors could strengthen this by providing a schematic with two sequential stages illustrating how neuroblast identity factors (STF code) change chromatin states to drive lineage-specific enhancer activation. The schematic can be shown from the neuroectoderm to individual NB lineages to make it more panoramic.

We thank the reviewer for this suggestion and for clearly articulating the conceptual point. As the reviewer points out, the literature has generally framed spatial-temporal integration as two alternative models—direct regulation at pre-accessible enhancers versus epigenetic priming by spatial factors. Our results suggest that elements of both mechanisms may operate within a lineage through a combinatorial STF code, with different spatial factors playing distinct roles (for example, Gsb contributing to chromatin priming, while En acts primarily at pre-accessible enhancers together with Hb). We agree that this central idea would benefit

from being illustrated more explicitly. In the revision we will add a schematic summarizing this proposed two-step model and clarify the relevant parts of the text.

(5) Identification of Priming Factors in NB7-4

While the authors suggest that an unknown priming factor might be responsible for establishing sites of integration in NB7-4, they do not identify or explore potential candidates for this role. Further investigation into what factors might be involved in chromatin priming in NB7-4 could provide a more complete understanding of the mechanisms at play.

We agree that identifying the factor responsible for establishing sites of integration in NB7-4 would be very informative. However, doing so would require substantial additional experiments to systematically test candidate spatial factors and assess their effects on chromatin accessibility in this lineage. Our goal in the present study was to establish how spatial and temporal cues are integrated at lineage-specific enhancers rather than to fully dissect all components of the STF code in each lineage. Identifying the priming factor in NB7-4 is therefore an important next step that we intend to pursue in future work, and we will clarify this point in the Discussion.

(6) Functional Validation of STF Code Components

The study proposes an STF code for each neuroblast lineage, but the specific components of these codes, beyond Gsb and En, are not fully explored. Identifying and validating additional factors that contribute to the STF code in each lineage could strengthen the conclusions.

We agree that identifying additional components of the STF codes operating in each lineage would be very informative. Our goal in this study was not to comprehensively define all spatial factors involved in each lineage, but rather to understand how spatial and temporal inputs are integrated at lineage-specific enhancers. By examining two well-characterized spatial factors with distinct properties – Gsb in NB5-6 and En in NB7-4 – we aimed to illustrate how different members of an STF code can play distinct roles in shaping chromatin accessibility and enhancer activation. Identifying additional factors that contribute to these lineage-specific codes will be an important direction for future work.

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